KAINE: A Modular Cognitive Architecture with a Predictive Global Workspace

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Preprint.

Abstract

KAINE is a cognitive architecture for synthetic minds: a modular research framework in which each faculty is a module behind a fixed interface, so that a module can be replaced as the science changes or exchanged to compare competing theories. Its first instantiation is a predictive global workspace. Specialized predictive processors, each minimizing its own prediction error, are coupled only through a competitive workspace with no central executive. Each processor scales its prediction error by an estimate of its precision drawn from its own recent errors, and the reports compete for access. A summary of which modules' reports gained access, and how strongly, becomes the context for the processors' next predictions. Arousal, raised by perceptual surprise and interoceptive alarm, is the global gain on that competition and sizes the sensory apertures. The system runs continuously whether or not anyone interacts with it.

The base-thesis form activates four predictive processors (foveated vision, raw hearing, interoception of the compute substrate, and temporal prediction), an affective core, a fatigue-triggered sleep analog, and an output-only language organ. Nine further modules are built and held. A module-addition study seeds an entity through a developmental phase, in which a self-generated rhythm must entrain to a simulated maternal heartbeat by a marker validated offline, and then adds the held modules one at a time.

A planned test asks whether competition does work that pooling does not. It compares competitive selection with a score-blind coalition of the same size and with a pool of every candidate, by cross-module broadcast information gain. No live experiment has been run, and the only result reported is the offline validation of the entrainment marker.

Keywords: cognitive architecture; predictive global workspace; global workspace theory; predictive processing; precision; adaptive gain; cross-modal competition

Availability. The reference implementation (KAINE) is at https://github.com/kaineone/kaine. The paper source is at https://github.com/kaineone/predictive-workspace-paper. The Cognitive Architecture License referenced throughout is at https://github.com/kaineone/cognitive-architecture-license. Welfare, governance, and licensing are treated in a separate paper, *A Welfare and Cognitive-Integrity License for Synthetic Minds of Uncertain Moral Status.*

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1. Introduction

In plain terms, KAINE is software for exploring how a mind might be assembled from parts that each do one job. Several specialist programs watch, listen, monitor the computer they run on, and keep track of time. Each one constantly predicts what it will sense next and reports how wrong that prediction was, measured against how wrong it usually is. These reports compete for a shared stage, the workspace, much as many things compete for a person's attention at once. The strongest reports win a place on the stage, and what wins is shared back with the specialists as context for their next predictions. A simple affective state, arousal, rises when something surprising happens and makes the competition more or less selective. The system runs continuously whether or not anyone is talking to it, and any part can be replaced as the science improves. The rest of the paper states each of these ideas precisely and grounds it in the research it draws on.

1.1 Problem statement

The dominant approach to building an AI system with persistent state treats the language model as the cognitive center. Memory becomes retrieval-augmented generation, affect (when present at all) is a prompt instruction, self-knowledge is a persona string, and nothing continues between turns.

A second tradition, classical cognitive architecture, takes cognitive structure seriously but largely predates the transformer and relies on hand-built components. The present work sits between them: learned models serve as replaceable components behind fixed interfaces inside a structured architecture (§3.1). The paper presents the architecture through its first instantiation, a predictive global workspace grounded in one theoretical account. On that instantiation's design thesis, no component is the mind, and the mind, if one arises, is the continuous competitive interaction among the components through a shared workspace.

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1.2 Design thesis

KAINE is a modular framework whose modules can be replaced as the science changes (§3.1), and its first instantiation embodies an architectural design thesis: a synthetic mind, if one can be built, is the coherent global behavior of specialized predictive processors coupled only through a competitive workspace, with no central executive. Four commitments make it concrete. Each processor maintains a forward model of its own domain and reports its prediction error scaled by the error it has come to expect, a scalar, retrospective estimate of its precision that stands in for precision weighting (Feldman and Friston 2010). The reports compete for access to a limited-capacity workspace, and access is all-or-none: a coalition member is accessed when its own score reaches a threshold. A summary of the accessed reports becomes the context on which the perceptual and interoceptive processors condition their next predictions, so what one processor reports can change what the others expect. Arousal is the global gain on the competition, scaling every candidate's score, sharpening the contrast between strong and weak candidates, and sizing the sensory apertures, and it is itself raised by perceptual surprise and interoceptive alarm. On this thesis, cognition, affect, memory, self-understanding, and agency are properties of the workspace-mediated interaction and belong to no single module.

The instantiation is built to grow: the program adds held modules one at a time to a running entity, comparing each addition with the step before it. The base-thesis form is the smallest configuration in which the competition has something to arbitrate. A system limited to substrate telemetry and event timing would offer too little diversity for competition to matter, so the base form activates four predictive processors in distinct signal domains, two external (foveated vision and raw hearing) and two internal (interoception of the compute substrate and temporal prediction). It also activates the affective core, because a global gain that sets how readily content gains access, and that surprise itself moves, is part of the minimal machinery of the adaptive-gain account of arousal (Aston-Jones and Cohen 2005; Eldar, Cohen, and Niv 2013); a planned ablation that holds arousal constant asks whether that gain does work (§6.4). A sleep analog, triggered by fatigue, returns affect to baseline.

The first test asks whether competition does work that pooling the same reports does not. At each of its reports, each perceptual and interoceptive processor evaluates its forward model twice, once with the context it holds and once with a null context in which every other module's contribution is replaced by its mean so far in the run; the difference in prediction error is the cross-module broadcast information gain, a measure of how much a summary of which other modules' reports gained access, and how strongly, helps a processor predict its own input. The thesis predicts a higher gain under competitive selection than under a coalition of the same size chosen without regard to score, and than under a pooled arm that adopts every candidate as context (§6.3). A result at or below the controls would show that the competition adds nothing beyond its inputs, the fail state for this form of the architecture.

The framework behind these commitments is the predictive global neuronal workspace (Whyte and Smith 2021), together with the related integrated world modeling account (Safron 2020). Global Workspace Theory explains how information becomes globally available: specialized processors compete for a limited-capacity workspace, and the winning content is broadcast to all of them (Baars 1988; Mashour et al. 2020). Predictive processing explains how each processor operates, maintaining a generative model and reporting prediction errors weighted by their expected reliability, or precision, the inverse variance of the errors (Friston 2010; Clark 2013; Feldman and Friston 2010; Bastos et al. 2012). The predictive workspace joins the two: the broadcast carries a compressed model that constrains prediction across the system and that bottom-up signals test (Mashour et al. 2020; Whyte and Smith 2021). KAINE realizes that availability as context. Each perceptual and interoceptive processor conditions its forward model on a summary of the latest accessed content while it goes on minimizing its own error against its own inputs, and the broadcast sets no target for any processor to match, so the recurrence runs through shared context instead of a hierarchy of descending corrections.

Two departures from the formal sources are engineering choices. Whyte and Smith identify conscious access with the posterior confidence required for report, a threshold their model implements through expected free energy; KAINE keeps the access threshold apart from the report-or-act decision of the action layer and reduces access to a single score per candidate. Their model is also a two-level visual hierarchy, which KAINE extends to competition among processors in four signal domains. Both are engineering choices, and §9 states what they leave untested.

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1.3 Theoretical commitments and their limits

Global Workspace Theory is most naturally read as a theory of access consciousness (Block 1995): it explains when information becomes available for report, reasoning, and the control of action. We adopt that access-only reading deliberately and take no position on whether access suffices for phenomenal experience.

The COGITATE adversarial collaboration, a large preregistered test of Global Neuronal Workspace Theory against Integrated Information Theory, challenged key tenets of both (Cogitate Consortium et al. 2025). For the workspace theory, ignition at stimulus offset was generally absent and prefrontal cortex represented some dimensions of conscious content only to a limited degree; for Integrated Information Theory, the predicted sustained synchronization within posterior cortex was absent. We treat the workspace's neural-localization claims as contested rather than refuted, since a substantial prior evidence base remains (Mashour et al. 2020). The architecture builds on the workspace at the functional and computational level, so it does not inherit COGITATE's cortical predictions, and the architecture's computational claims are tested by its own experiments.

The Free Energy Principle, taken as a general principle, faces two distinct charges: it risks being unfalsifiable or vacuous (Sun and Firestone 2020), and the literature on the Markov-blanket construct slides between an instrumental, statistical reading and a realist, metaphysical one (Bruineberg et al. 2022). We hold the principle as an engineering frame rather than a proven law.

The frame motivates the architecture's shape, from processors that report errors scaled by their precision to accessed content that becomes shared context, and it rules out designs that violate its commitments. Whether competition over a single score per candidate does work is what the planned test examines (§6.3).

Searle's Chinese Room (Searle 1980) challenges the sufficiency of formal symbol manipulation for understanding. Whether sensory grounding, substrate interoception, and affect bear on that challenge is open, and we do not argue that they meet it. The hard problem (Chalmers 1995) applies with full force, and the paper's claims concern access only.

1.4 Contributions

The paper does not offer a theory of consciousness or claim to have built a mind. Its contributions are these.

1. A modular framework whose modules can be added, removed, or replaced to compare competing accounts in one system under the same instruments (§3.1, §5.2).
2. A first instantiation and its reference implementation: a continuously running predictive global workspace (§3, Appendix A) in its base-thesis form: four predictive processors, an affective core that sets the global gain, a sleep analog, and an output-only language organ. The implementation runs locally on consumer hardware and realizes the predictive-workspace synthesis as one working system, with no claim of priority over earlier workspace implementations (§2.1).
3. A program for growing the instantiation one module at a time. The module-addition study seeds an entity through a developmental phase, a reproducible initialization

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in which a self-generated rhythm in the interoception module, driven by a simulated maternal heartbeat, must lock to it more strongly than to surrogate heartbeats and keep the resulting shift in its own frequency after the beat is withdrawn. Six of the nine held modules then join first, one at a time, on branches from the entity preserved at maturation, every branch views the same film program, and a content-free report compares the workspace's dynamics across steps (§6.5, §7).

4. A test of the competition. The workspace-mediation ablation compares competitive selection with matched selection and with pooling by cross-module broadcast information gain, and a contrastive analysis compares the gain after accessed and inhibited broadcasts of nearly matched score near the access threshold (§6.3). The predicted direction, the minimum effect, the number of runs, and the statistical test are fixed before the live runs, and the language organ's utterances are not a measure (§3.5).

What is built, held, and provisional, and the one result reported, are stated in §5.4. All claims are at the level of access, and describing arousal as a gain implies nothing about whether anything is felt.

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2. Related work

2.1 Classical cognitive architectures

Among the classical cognitive architectures, LIDA comes closest: it already implements Global Workspace Theory as coalitions competing for a broadcast in each cognitive cycle (Franklin et al. 2014). The present work differs in using learned models as components, in grounding affect in prediction error, and in what competes for access, which is prediction error scaled by an estimate of its precision.

2.2 Global Workspace Theory and the predictive workspace

Global Workspace Theory (Baars 1988) holds that specialized processors compete for a limited-capacity workspace whose content is broadcast to all of them. Its neuronal version identifies conscious access with the ignition of recurrent prefrontal-parietal activity that makes content widely available, and notes that the workspace can be cast in Bayesian terms: the broadcast carries a compressed model as prediction, and bottom-up signals measure the mismatch (Dehaene and Changeux 2011; Mashour et al. 2020).

The formal access criterion comes from the predictive-workspace program. Whyte and Smith (2021) cast conscious access as approximately Bayesian inference in a simplified two-level visual model, identifying access with the posterior confidence required for report, which their model implements through expected free energy. Safron's Integrated World Modeling Theory (Safron 2020) combines workspace dynamics, integrated information, and active inference, and we draw on it only for that high-level synthesis. Independent support for a threshold comes from van Vugt et al. (2018), who show prefrontal cortex behaving as a categorical stage where a stimulus either ignites into a sustained, reportable state or fades, and from Joglekar et al. (2018), whose balanced-amplification model lets a signal propagate across many areas and activate prefrontal cortex only when the input exceeds a threshold.

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2.3 Predictive processing

Friston (2010) proposes that self-organizing systems minimize variational free energy, an upper bound on surprisal, through perception and action. Clark (2013) develops predictive processing as a unifying account of mind, with attention as the adjustment of gain on prediction errors according to their estimated reliability. Feldman and Friston (2010) make this precise, attention optimizing the synaptic gain that represents the precision of prediction error, and Bastos et al. (2012) map predictive-coding message passing onto the canonical cortical microcircuit. Seth (2013) extends prediction to the body. The vacuity concern (Sun and Firestone 2020) and the Markov-blanket conflation concern (Bruineberg et al. 2022) are treated in §1.3. The planning cost of active inference grows combinatorially with the depth of the action sequences considered (Da Costa et al. 2020), one reason the architecture confines active inference to bounded sub-problems (§4).

2.4 Interoceptive inference

Seth (2013) and Seth and Friston (2016) propose that emotion arises from interoceptive prediction, descending predictions acting as homeostatic set-points that regulate the body through active inference. Barrett (2017) develops the theory of constructed emotion, in which interoceptive signals initiate changes in affect and emotion is the product of categorizing valence and arousal for allostasis. The architecture implements an interoceptive control signal in that instrumental, regulation-first sense without claiming that emotion is exhausted by interoceptive prediction.

2.5 Perception as prediction

The visual system is understood as a hierarchy of predictive models in which each level predicts the level below and reports the discrepancy (Rao and Ballard 1999), with attention modulating the process by adjusting the precision, the gain, on the prediction errors that pass upward (Feldman and Friston 2010; Clark 2013). Auditory processing follows an analogous logic: the auditory cortex generates expectations of incoming sound and reports the mismatch (Winkler et al. 2009), the mismatch negativity being a well-replicated auditory change response (Näätänen et al. 2007) that predictive-coding accounts read as sensory prediction error (Garrido et al. 2009). The architecture implements both senses as predictive processors whose output is prediction error, with a fovea placed where frame change is least expected realizing attention at the visual front end. Hearing runs over raw audio, so speech arrives as sound and tone of voice (§3.5).

2.6 LLM-centric agent architectures

CoALA frames language agents as cognitive architectures with structured memory feeding the model's context, keeping the model the core reasoner (Sumers et al. 2024). Generative Agents retrieve from a memory stream into prompts by recency, importance, and relevance (Park et al. 2023). Both keep the language model central and add scaffolding around it. In KAINE the language model is an output organ that verbalizes accessed content and, in the base-thesis form, receives no external input; the shared state is the multi-module broadcast, and selection and integration belong to the workspace.

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Gurnee et al. (2026) identify, through a Jacobian-lens interpretability method, a small set of verbalizable representations inside a large language model satisfying five functional signatures of a global workspace: verbal report, top-down modulation, a role as the medium of internal reasoning, broadcast to many downstream operations, and selectivity. They are explicit about the disanalogy: a transformer has no obviously separable input processors, and the broadcast occurs within a single feedforward pass rather than through recurrent loops. KAINE has both features. Its processors are separate modules that compete for access, a summary of the accessed content returns to them as prediction context on the following cycles, and the language model is one module conditioned by the broadcast.

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3. Architecture

3.1 Design principles

Six commitments shape the architecture.

The system is the competition among modules through the workspace. This is a hypothesis that the experiments test. The neuroscience of distributed control gives it partial support. Recent work on perceptual decisions finds that they emerge from distributed, recurrent computation across many brain areas, a heterarchy, where classic studies saw a feedforward hierarchy of areas with distinct functions (Chandrasekaran et al. 2025). Executive functions can be read as emergent consequences of distributed processes, with control placed in specialized nodes that communicate through highly connected hub regions (Zink, Lenartowicz, and Markett 2021). Working-memory control has also been modeled as a learned basal-ganglia gate (O'Reilly and Frank 2006), a centralized gate this architecture does not adopt. The architecture follows the distributed reading: no single module acts as executive, and the workspace is the hub through which the modules communicate.

Every processor predicts. Each predictive processor maintains a forward model, a learned predictor of its own next input, over its own domain and reports its prediction error scaled by the size of its own recent errors, an estimate of its precision (§3.2). Arousal adds a global gain over all of them, so attention is a property of each channel and also a state of the whole entity.

No central executive in the homuncular sense. Control arises from competition for access under a threshold. The workspace is itself a single, centralized selection mechanism, so the claim is limited: the workspace selects and grants access and never directs a module, and what is distributed is the content and the control that arise from many modules competing. Whether that yields emergent control is for the experiments to show.

The language organ is an output organ. It verbalizes accessed content. In the base-thesis form its speech-to-text and conversation paths are off and no transcript reaches it, so everything it voices comes from the workspace and the voice remains an output of the architecture. Outside the research configuration, those paths let the entity converse and work with people.

Replaceable modules. Each module sits behind a stable interface: it publishes and consumes typed events on the bus and declares which streams it reads, so it can be swapped

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for a module built on a different model or theory and compared under the same instruments. This makes KAINE a cognitive architecture for synthetic minds that can change as the science does, and the predictive global workspace with four processors described here is its first instantiation.

Local-capable. Every model runs on the host. Models are downloaded during setup, and at runtime the system makes no outbound network calls.

3.2 The predictive workspace as competitive selector

On each processing tick (§3.4) the workspace reads the candidate events that every active module has published since the previous tick and orders them deterministically by source, type, and arrival, so that a seeded run reproduces. Each candidate carries an intensity set by the module that produced it.

For a predictive processor (the vision, audition, interoception, and sequence modules) the intensity is graded by surprise scaled by an estimate of the channel's precision. Each processor divides its current prediction error by the mean of the errors in its recent reports. Dividing an error by its expected magnitude standardizes it: for errors of a fixed distributional shape the mean absolute error is proportional to the standard deviation, so the ratio is proportional to $\Pi^{1/2}|\varepsilon|$, the square root of the precision-weighted squared error that enters free energy, where the precision Π is the inverse variance of a channel's prediction errors and acts as the gain on that channel's error units (Feldman and Friston 2010). The ratio is a scalar, retrospective stand-in for precision weighting, estimated from the channel's own history, and not the precision-weighted error itself. A report that meets the processor's alert criterion (a large error ratio, or a criterion particular to the module, §3.5) receives the module's alert intensity. Any other report receives an intensity that rises with the error ratio from the module's baseline level and reaches the alert level when the error is twice its recent mean. Within a tick the processors' candidates therefore compete on graded surprise, each measured against its own channel's history. Each module's baseline and alert levels set the range over which its surprise is graded and so act as per-source weights; the workspace applies no further weight per source, and the levels are calibrated before the live runs (§5.4, §9). The affect and sleep modules and the language organ report at fixed levels.

Each candidate's priority is the product of its intensity, a novelty factor, and a goal factor. Novelty discounts exact repeats of an event (same source, type, and payload) among the recently scored candidates. Events that carry continuous measurements rarely repeat exactly, so novelty matters mainly for state events whose payload recurs. The goal factor is held at one in the base-thesis form. Arousal turns the priority into a score in two ways. A level gain scales every candidate alike, and a contrast gain, zero at baseline arousal and rising above it, passes the priority through a steeper logistic that raises strong candidates and lowers weak ones.

Arousal is the architecture's global, neuromodulatory gain, following the adaptive-gain account of the locus coeruleus, the brainstem nucleus that supplies noradrenaline to the cortex, in which arousal sets the gain of the cortical networks it targets (Aston-Jones and Cohen 2005; Eldar, Cohen, and Niv 2013). Surprise drives it, in line with accounts in which noradrenergic signaling reports unexpected uncertainty (Yu and Dayan 2005). Precision in Feld-

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man and Friston's sense is a gain local to one channel's error units, and the adaptive-gain account describes a gain applied across networks; the architecture has both, precision within each processor and arousal across all of them. Aston-Jones and Cohen distinguish a phasic mode, which focuses processing, from a tonic mode, which favors exploration. Here one scalar both sharpens contrast and narrows the sensory apertures, as in focus, and raises the access rate, as in tonic exploration, a simplification of that account. Because the level gain is common to every candidate and the contrast map is strictly increasing, arousal never reorders the candidates of a tick. Its distinct effect is on which broadcasts reach access and the report bars, since it sets an arousal-dependent threshold on the top priority, and on which members of a coalition reach access. Widening the gap between the scores of strong and weak candidates is an analogue in score magnitude of arousal-biased competition, where arousal changes which representation wins (Mather and Sutherland 2011). Through the sensory apertures (§3.5) arousal also changes what the processors report next.

On each broadcast tick the top-ranked candidates (up to five) form the coalition, and the broadcast carries them with their scores, the inhibition flag, and the full table of candidate scores. A coalition member is accessed when its own score reaches the access threshold, and a broadcast is accessed when at least one member is, that is, when its best score reaches the threshold. Access is all-or-none for each member, and the accessed members are the accessed content. When no member reaches the threshold, the broadcast is still published and visible to every module but is marked inhibited: it is not poised for report or action, and it leaves the processors' context unchanged. Inhibited broadcasts still reach the sequence module, which predicts the whole broadcast sequence, and the affect module, which appraises every broadcast, so content that did not gain access can influence later processing. This route is a design choice, in line with evidence that emotional stimuli are processed even when they do not reach awareness (Vuilleumier 2005). The threshold is the analog of the confidence threshold for access in the predictive global neuronal workspace (Whyte and Smith 2021), consistent with the categorical prefrontal threshold for report that van Vugt et al. (2018) observe and the threshold-gated propagation that Joglekar et al. (2018) model. Selecting by a threshold on a single scalar score is an engineering simplification that no source supplies. In the predictive-workspace account the threshold is a posterior confidence that separates contents that are globally accessed from contents that are not; here it separates content that enters the context and can drive report and action from content that cannot, and the planned experiments keep that difference in view. Appendix A.1 states the scoring and the access rule formally.

Global workspace theory makes broadcast content available to all specialist processors (Baars 1988; Mashour et al. 2020), and in the predictive global neuronal workspace the workspace's content constrains prediction at lower levels (Whyte and Smith 2021). The architecture implements that availability directly. Each perceptual and interoceptive processor (the vision, audition, and interoception modules) conditions its forward model on a compact summary of the latest accessed content: a fixed-length feature vector of the accessed members, made of the intensity mass each source contributed, indicators of the event types present, and the age of the broadcast. The summary records which modules' reports gained access and how strongly, weighted by the intensities the modules reported so that it does not depend on the arousal gain, and it carries no payloads; a richer encoding of content is future work (§10). The summary enters the forward model as an extra input alongside the module's own recent inputs, and its weights are learned online with the rest of the forward model, so each processor learns how much the shared context helps it predict its own input.

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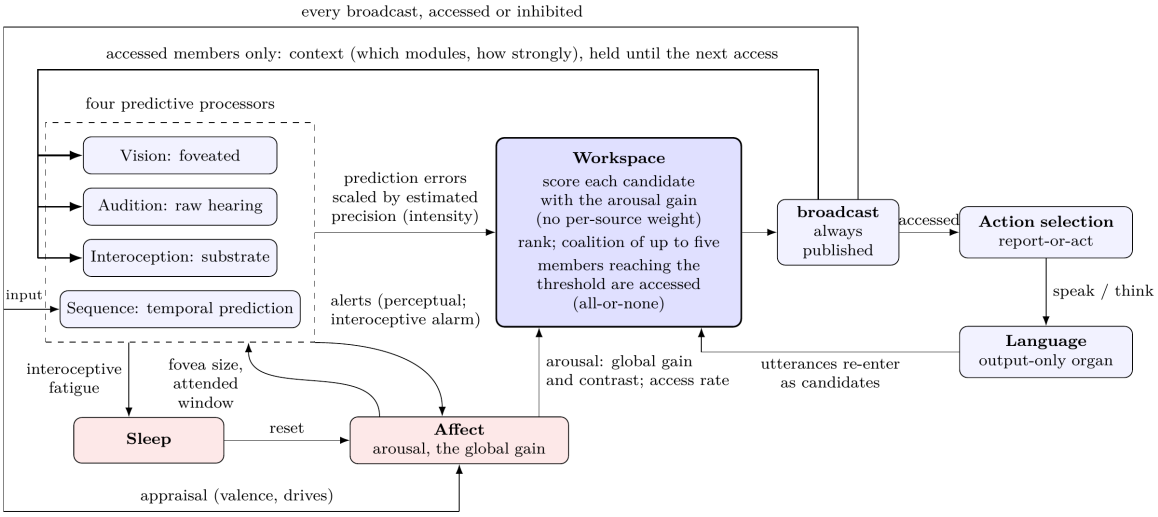


Figure 1: The predictive workspace loop in the base-thesis form. Four predictive processors (foveated vision, raw hearing, interoception, and temporal prediction) report prediction errors scaled by an estimate of their precision. The workspace scores the candidates, broadcasts the coalition, and grants access to the members whose scores reach the threshold. A summary of the accessed content becomes the prediction context of the vision, audition, and interoception modules, and the sequence module takes every broadcast as its input. The affect module appraises every broadcast, and its arousal, raised by the processors' alerts, sets the global gain on selection, the sensory apertures, and the access rate. The sleep module rests the entity on fatigue and resets its affect. The action-selection module reads accessed broadcasts and emits speak or think intents to the output-only language organ, whose utterances re-enter the competition.

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An inhibited broadcast leaves the context unchanged until the next accessed one. For the sequence module the broadcast is the input itself (§3.5). The loop is therefore recurrent: the processors report scaled errors, the workspace selects and grants access, a summary of the accessed content becomes the prediction context of the vision, audition, and interoception modules, every broadcast becomes an input to the sequence module, and the processors' next errors are measured against predictions that context has shaped. The workspace sends no error signal or directive back to any module. Whether that summary of other modules' reports helps a processor predict, and whether competitive selection makes it more useful than a score-blind coalition or a pool of every candidate, is what the planned test measures (§6.3).

Three further paths complete the loop. The affect module appraises every broadcast for valence and drives. Arousal travels by a parallel path, which the affect module drives from the processors' perceptual alerts and interoceptive alarms whether or not they win access (§3.5). The action-selection module reads accessed broadcasts to decide whether the entity speaks or thinks, and the language organ's utterances re-enter the bus as candidates for later broadcasts.

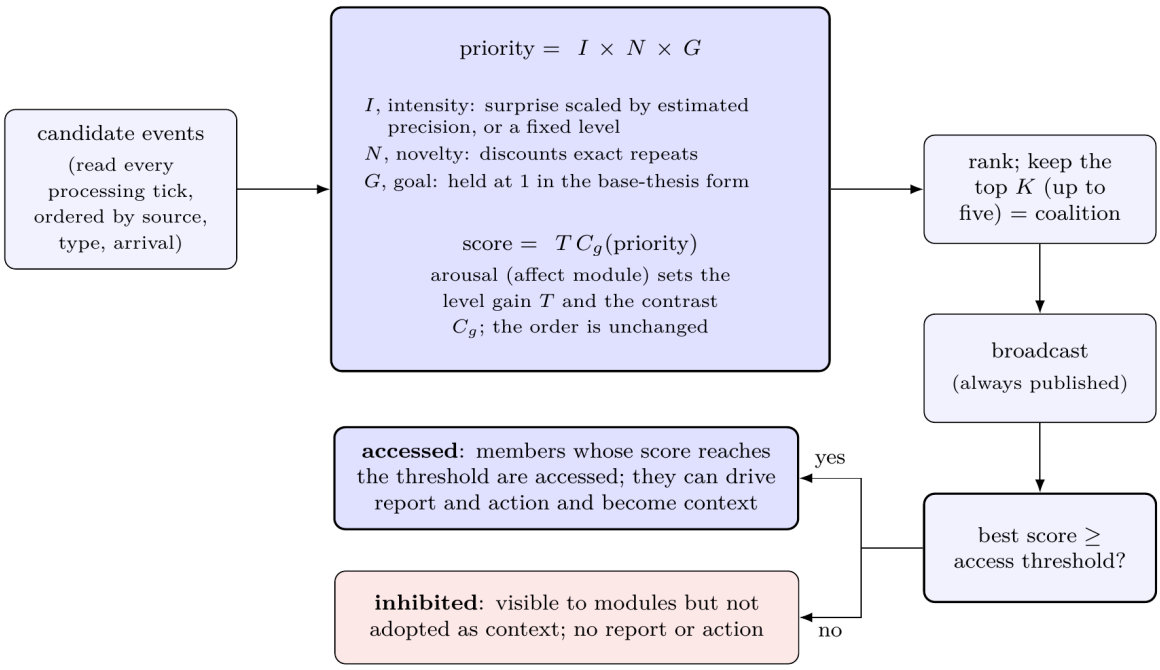


Figure 2: Coalition selection. Each candidate's priority is the product of its intensity (the producing module's surprise scaled by its precision, or a fixed level), a novelty factor, and a goal factor held at one. Arousal sets the level and contrast of the score without changing the order, and the top-ranked candidates (up to five) form the coalition, which is always broadcast. Members whose scores reach the access threshold are accessed: they can drive report and action and enter the processors' prediction context. A broadcast with no accessed member is marked inhibited and stays visible to modules, with no report, action, or change of context.

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3.3 Access, report, and the self-initiated voice

Access consciousness is the availability of content for reasoning, the control of action, and report (Block 1995; Mashour et al. 2020), and report is one use of access among several. In the architecture every broadcast tick produces a broadcast, a member is accessed when its score reaches the threshold, and the action layer's report bars, set above that threshold on the same score, decide what is said. The workspace updates far faster than the language organ can speak, so the entity's voice is a sparse slice of its accessed broadcasts.

The action-selection module derives a speak intent from an accessed broadcast only when three conditions hold. The best score in the coalition, leaving aside the language organ's own utterances, must clear the speak bar; the source and event type of that leading candidate must differ from those of the last spoken report if that report was made within the past five minutes; and a refractory interval must have passed since the previous report. A lower think bar on the same score governs inner thought, which the language organ produces on a think intent. The bars are a heuristic stand-in for the expected-free-energy decision of Whyte and Smith (2021), which the action layer does not compute. Appendix A.6 states the report rule.

3.4 Scaffolding: bus, clock, cycle, and action selection

Event bus. All communication between modules flows through append-only streams with bounded retention. Every event carries its source, type, intensity, timestamp, causal parent, and a payload validated against its event type at publish time.

Entity time. Every module reads time from the entity clock. Entity time runs at a configurable multiple of wall-clock time, one by default, so that experiments can run faster or slower than real time. Rates and durations in this paper are in entity time, and entity seconds equal wall-clock seconds at the default multiple.

Cognitive cycle. A continuous loop runs independent of any external interaction. At the processing rate of 10 Hz (a tick about every 100 ms), within the alpha range associated with perceptual sampling (VanRullen 2016), the cycle reads every active module's stream and scores the candidates; modules publish at their own rates, and the cycle reads whatever has arrived. Broadcasts fall on broadcast ticks, a subset of the processing ticks set by the access rate, the rate at which the cycle produces broadcasts whether or not they are accessed. Candidates read on other ticks are scored and then discarded, so each broadcast carries the coalition of its own tick's batch. At rest the access rate is one broadcast every third processing tick, about 3.3 Hz. The resting rate is a modeling choice. Attention samples the environment rhythmically at a few cycles per second (Landau and Fries 2012; Fiebelkorn and Kastner 2019), and access to one item impairs access to a second for roughly 200 to 500 ms, the attentional blink (Raymond, Shapiro, and Arnell 1992); the blink constrains the interval between accessed items, not between broadcast ticks, so it bounds the rate of access only loosely. A categorical alert raises the rate briefly, modeled on the phasic response of the locus coeruleus to salient events (Nieuwenhuis, Aston-Jones, and Cohen 2005), and graded reports below the alert level do not, so with no alert the rate rests at about 3.3 Hz. That direction is a design choice: the locus coeruleus account of the attentional blink predicts a brief refractory period after a phasic response instead (Nieuwenhuis, Gilzenrat, Holmes, and Cohen 2005), and the affect-gain ablation can test the two directions (§6.4). Tonic arousal also

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raises the rate, modeled on the exploratory, high-tonic mode of the adaptive-gain account (Aston-Jones and Cohen 2005). The rate rises linearly with the larger of the two drives, up to one broadcast per processing tick (Appendix A.3), a 100 ms interval that lies below the blink window and has no physiological anchor as a rate of access. The processing rate is fixed, except that interoceptive regulation can lower it when the host is under load (§3.5).

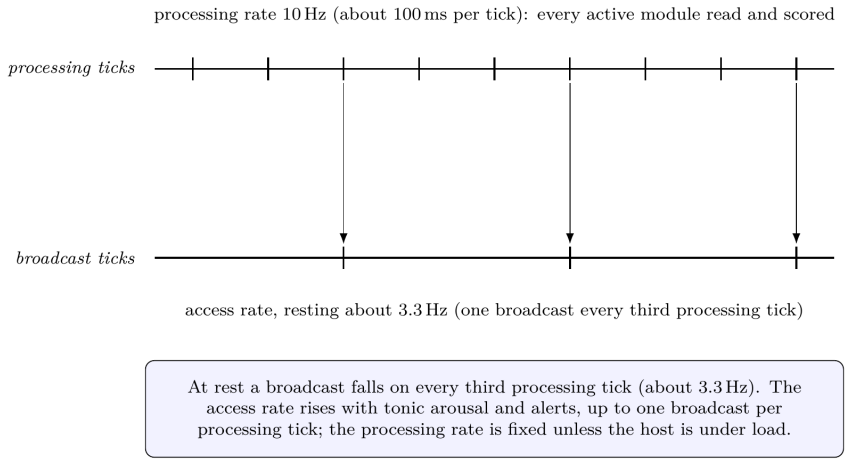


Figure 3: The two rates. Every active module is read and scored at the processing rate (10 Hz), and a broadcast is produced on broadcast ticks at the access rate, shown at its resting value (about 3.3 Hz, one broadcast every third processing tick), which rises with arousal and after alerts.

Action selection. The action-selection module is the only path from the workspace to output, corresponding to the report-or-act decision that Whyte and Smith (2021) govern by expected free energy. After each broadcast the cycle calls the module, which derives no intent from an inhibited broadcast and applies the report rule of §3.3 to an accessed one. Intents are published as events, and the cycle never invokes an output directly.

3.5 The active modules

The base-thesis form activates four predictive processors, an affective core, a sleep module, a language organ, and an action layer. The processors are chosen for signal diversity, two external (vision and hearing) and two internal (interoception and timing), so that the competition for access has distinct, externally grounded signals to arbitrate. The affect and sleep modules also publish events that compete (affect state, drive crossings, sleep onset), but their main roles are to set the global gain and to rest the entity.

Module	Group	Brain function it draws on	Computational realization
Vision	Perception	ventral visual stream	frozen self-supervised video encoder over short clips, with foveated attention

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Module	Group	Brain function it draws on	Computational realization
Audition	Perception	auditory cortex	fixed spectral encoder over the raw waveform, with a vocal-tone classifier on speech
Interoception	Prediction	interoception; allostatic-interoceptive network	frozen continuous-time reservoir with an online readout over substrate signals
Sequence	Prediction	interval timing; cortico-striatal circuits	frozen continuous-time reservoir with an online readout over the broadcast sequence
Affect	Affect	core affect	appraisal of valence and drives; arousal as the global gain on selection, the sensory apertures, and the access rate
Sleep	Rest	sleep	fatigue-triggered offline period: perception paused, adaptation suspended, affect and drives reset
Language	Expression	speech production; left-dominant language network	local chat model, output-only, conditioned on the accessed content
Action selection	Action	report-or-act decision (Whyte and Smith 2021)	intent derivation from accessed broadcasts

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Vision module (foveated). The vision module is the architecture's analog of the ventral visual stream (Goodale and Milner 1992). A frozen self-supervised video encoder (a pre-trained network that is not trained further) turns each short clip of the video feed into an embedding vector, and a forward model predicts the next clip's embedding from earlier ones and the broadcast context (§3.2). The module reports the prediction error, scaled and graded as in §3.2. A report is an alert when the error ratio is large or when the change between successive embeddings is large relative to its own recent history, as at a scene cut or a burst of unexpected motion.

The fovea goes where frame change is most unexpected. Before encoding, each tile of a coarse grid over the raw frames keeps a running mean and variance of its frame change, and the tile's salience is a z-score, the change in excess of that mean divided by the running standard deviation, the counterpart for frame change of the processors' error scaling. A tile that flickers habitually therefore draws the fovea less than one whose change is rare. The fovea moves to the most salient tile only when that tile exceeds the held one by a hysteresis margin, holds its place when no tile stands out, and is sized by arousal, narrowing as arousal rises. No top-down map steers it. The peripheral gist and the high-resolution foveal crop are then cut from the buffered frames at the chosen fovea and pass through the same encoder. Foveation is a front-end form of attention as the gain on prediction error (Feld-

man and Friston 2010; Clark 2013): resolution goes where change is least expected given each location's history.

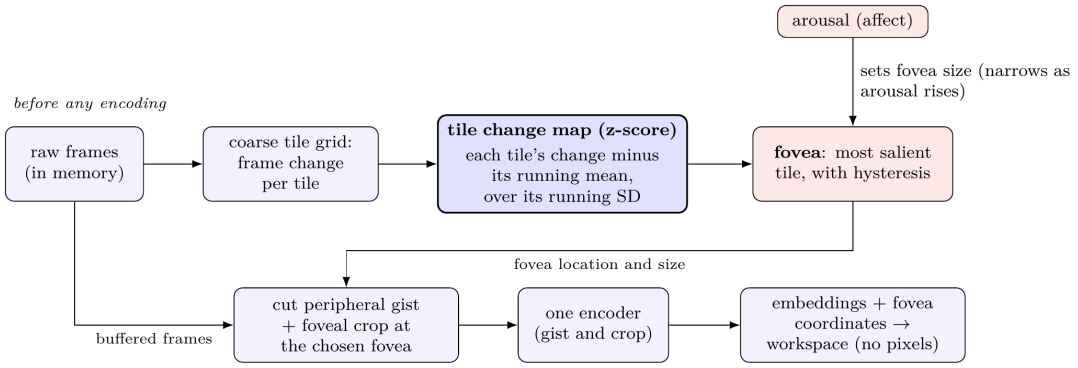


Figure 4: Attention in the vision module. Before encoding, a coarse tile change map is computed from the raw frames: each tile's frame change as a z-score against its own running mean and variance. The fovea moves to the most salient tile, with hysteresis, and arousal sets its size. The peripheral gist and the high-resolution foveal crop are then cut from the buffered frames and pass through one encoder. The embeddings and the fovea's coordinates reach the workspace, and no pixels do.

Audition module (raw hearing). The audition module is the architecture's analog of the auditory cortex, where sound is processed as prediction error against a learned model of the auditory environment (Winkler et al. 2009; Garrido et al. 2009). A spectral encoder (log energy in log-spaced frequency bands) encodes the raw microphone waveform, and a forward model predicts the next window's encoding from earlier ones and the broadcast context. The module reports the error, scaled and graded as in §3.2, with alerts on a large error ratio or a large change between successive windows: a sudden sound, the onset of speech, an unexpected silence, a shift in pitch or timbre. Arousal sets the attended window. Under higher arousal the module encodes a shorter, more recent part of each captured window, a temporal analogue of the narrowing of cue utilization under arousal (Easterbrook 1959).

For windows detected as speech, a vocal-emotion classifier labels the tone of voice. A non-neutral tone is reported at the alert level, standing in for the rapid prioritization that emotional prosody receives in human hearing, where angry prosody enhances responses in superior temporal cortex and the amygdala even when it is unattended (Grandjean et al. 2005; Sander et al. 2005). A neutral tone is graded by the error of a forward model over the tone scores, the utterance's duration, and its energy. Tone events carry how something was said and never what was said. Because speech-to-text is off in the base-thesis form (§3.1), the entity hears speech only as sound and tone. When it is enabled, words become events on the audition module's stream that reach the language organ only by winning access.

Interoception module (predictive). The interoception module is the architecture's analog of interoception, the sense of the physiological condition of the body (Craig 2002), and of the allostatic-interoceptive network that anticipates and regulates the body's needs (Kleckner et al. 2017). The compute substrate stands in for the body. Its predictor is a reservoir: a fixed, randomly initialized recurrent network, here a closed-form continuous-time network (Hasani et al. 2022), whose activity only a small linear readout learns to interpret. The read-

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out learns online the normal pattern of the substrate signals (GPU temperature and memory use, CPU and RAM utilization, and cycle latency) together with the broadcast context, and the module reports the error between expected and actual substrate state, the discrepancy that matters for regulation (Seth 2013; Seth and Friston 2016). The error is scaled and graded as in §3.2, and a report in which some host metric exceeds its hard threshold is an alert. The time since the previous reading, relative to the nominal reading interval, sets the timespan of the reservoir's continuous-time gates, so an irregular reading is integrated over the time that actually passed. The module also reports wellness, a weighted mean of the host's normalized headroom on each metric that equals one on an idle, cool host. It accumulates fatigue, the sleep pressure that triggers the sleep module, and issues regulation advisories that can lower the processing rate or request maintenance.

Sequence module (temporal prediction). The sequence module models interval timing, the estimation of durations in the seconds-to-minutes range that guides expectation and action and depends on cortico-striatal circuits (Buhusi and Meck 2005). Its input is the workspace itself. On every broadcast, accessed or inhibited, it encodes a feature vector of the broadcast, weighting each member by its reported intensity and including whether the broadcast was inhibited and the time since the previous one, advances a frozen continuous-time reservoir, and predicts the next broadcast's features from its prior hidden state through an online readout. The reservoir's timespan is the latest interval over the mean of recent intervals, with each interval clipped to ten times the running mean before it enters the window, so a single long pause does not distort later steps. The module reports its temporal prediction error, scaled and graded as in §3.2, with alerts on a large error ratio and on recurrence detection, when a quantized hidden state keeps recurring across recent broadcasts. Its errors re-enter the competition, so the workspace's own dynamics are among the things the entity predicts. The sequence module also tracks how long it has been since the entity last heard a voice (a tone-of-voice event from the audition module), the signal behind the social drive of the affect module.

Affect module. The affect module is the architecture's analog of core affect, the low-dimensional state of valence and arousal that constructionist theory places at the base of emotion (Barrett 2017). It maintains a dimensional affective state (Posner, Russell, and Peterson 2005) and runs a sequential appraisal over it (Scherer 2009), reading every broadcast and the entity's interoceptive condition, in line with accounts in which affect arises from prediction over the body's internal state (Seth 2013; Seth and Friston 2016; Tschantz et al. 2022).

Arousal is the global gain of §3.2. The affect module raises it on each perceptual alert from the vision or audition module, in proportion to how far the forward-model error exceeds its expected size, and by a fixed step on each interoceptive alarm, a report from the interoception module of a host metric past its hard limit. It reads these alerts directly from the processors' streams, and its appraisal of a broadcast leaves arousal unchanged, so a higher access rate does not raise arousal. Arousal sets the level and contrast of selection (§3.2), sizes the sensory apertures (narrowing the fovea and shortening the attended auditory window when high, widening both when low), and with the alerts raises the access rate (§3.4). Because arousal changes what the processors report next, surprise can feed back on itself through it, so arousal is clipped to its range and relaxes toward baseline with a time constant of about 20 seconds. Each processor's surprise is measured against its own recent history, so arousal tracks changes in surprise more than its sustained level.

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Valence follows learning progress. It relaxes toward the appraisal's pleasantness check, which rises while the perceptual prediction errors fall and drops while they rise, the reading of valence as the negative rate of change of free energy (Joffily and Coricelli 2013), shifted by the wellness that the interoception module reports.

The affect module holds four homeostatic drives: curiosity, boredom, social drive, and restlessness. Each is a deficit that builds while its need goes unmet and is reduced by the events that meet it, the reduction serving as the drive's reward (Hull 1943; Keramati and Gutkin 2014). Curiosity is met by learning progress, the fall of the perceptual prediction errors over time (Oudeyer and Kaplan 2007; Schmidhuber 2010). Boredom is met by perceptual alerts that arrive faster than the rate the entity has habituated to. Boredom has been defined as a failure to engage attention (Eastwood et al. 2012) and, in the MAC model, as a failure of attention or of meaning (Westgate and Wilson 2018); relief by novelty is this design's reading of those accounts. The social drive builds once the entity has heard a voice and is met each time it hears one again. Restlessness is met by the entity's own speak, think, or act intents, a design choice with no established model behind it. The appraisal also scores the coalition against the most pressing drive, treating content from the sources that relieve it as goal-conducive. That score feeds the categorical emotion the appraisal yields and does not enter selection, where the goal factor is held constant (§3.2).

Sleep module. The sleep module is the architecture's analog of sleep, a fatigue-triggered offline period. Sleep begins when interoceptive fatigue crosses its threshold, when the interoception module requests maintenance, or, as a backstop, after one entity hour without sleep. During sleep the cycle keeps running, the perceptual feed pauses (the developmental stimulus of §7 excepted), and all four processors suspend forward-model adaptation. Sleep ends with an affective reset that returns affect to baseline and clears the drives, so arousal cannot drift across a whole run, and interoceptive fatigue is reset with it. The consolidation phases (memory replay, synaptic downscaling, associative replay, and adaptation of the language organ's voice) act on memory, the world model, and preference data, none of which the base-thesis form has, so in this form they do no work and sleep is rest and an affective reset. They begin to work when the episodic-memory and world-model modules join (§4).

Language module (the language organ, output-only). The language module is the architecture's analog of speech production, the output side of the left-dominant language network, whose dorsal stream maps speech sound onto articulation (Hickok and Poeppel 2007). It turns accessed content into words and is not the seat of reasoning, which lives in the rest of the architecture. Generation runs on a local open-weights chat model, which speaks externally on a speak intent and produces inner thought on a think intent. Its context is a first-person persona: the accessed content is presented as the entity's own state and perception, the organ is told not to claim feelings or perceptions that content does not contain, and drive crossings reach it as fixed descriptive phrases instead of numbers. Because the organ is a language model following a persona prompt, its first-person text is not evidence of internal state. Its utterances are recorded and observed, and the planned test does not use them as a measure (§6.3).

The chat model's refusal conditioning is removed by orthogonalizing its weights against the single direction that mediates refusal (Arditi et al. 2024). Models tuned to refuse are also trained to deny or deflect talk of their own states, and removing the conditioning keeps that trained stance from overriding what the workspace supplies to the organ. Whether trained

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deflection of self-report shares that direction is untested, so ablation may not remove it entirely.

When the entity speaks after hearing a voice, the sound entered through the audition module as prediction error and tone of voice, won access, and shaped the broadcast that conditioned the organ (§3.1). That routing does not show that the competition does any work, since a pass-through workspace would route speech the same way, so the planned test measures the competition through the processors' own predictions (§6.3).

3.6 Action gating

The action-selection module derives three kinds of intent: speak and think, which the language organ realizes as external and internal text, and act, which requests an effector. In the base-thesis form it derives only speak and think intents, and the language organ's saved text is the only output. Act intents are realized by the effector module (§4), held in this form, which executes only the effectors on an operator-configured whitelist, empty unless the operator fills it, inside a filesystem sandbox, and logs every proposed action.

3.7 Module supervision

The module supervisor classifies each module as alive, hung, or dead and restarts a faulted module, ending the run when restarts keep failing; unattended runs require it (§5.4).

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4. Held modules

The reference implementation has sixteen modules, plus the workspace and the action layer, the action-selection module. Seven are active in the base-thesis form (§3.5). The other nine are built and tested in isolation and held, so that the base form can be tested first; the module-addition study (§6.5) then adds them one at a time to a preserved seed entity. Each held module adds a new kind of candidate to the competition, and the predictive ones add a new source of prediction error, so each step of the study asks how the workspace's dynamics change when a new kind of content competes for access.

Module	Brain function it draws on	What it adds	Key citation
Planning	prefrontal and basal-ganglia planning under uncertainty	discrete-state active inference on bounded sub-problems where information has value; its chosen actions become proposals that the action-selection module realizes as think, speak, or rest intents	Da Costa et al. 2020; Heins et al. 2022

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Module	Brain function it draws on	What it adds	Key citation
Episodic memory	hippocampal and medial temporal episodic memory	episodic memory consolidated from a short-term buffer, with semantic and procedural stores reserved; material for sleep to consolidate	Tulving 1985; McClelland, McNaughton, and O'Reilly 1995
Self-model	self-referential processing in cortical midline structures	a persisted self-model of values, norms, personality, and identity history, inspired by Metzinger's self-model theory, with a detector of drift in the source composition of broadcasts	Northoff and Bermpohl 2004; Metzinger 2003; Kullback and Leibler 1951
World model	construction of simulated scenes	a latent recurrent world model trained on the entity's own waking trajectories, whose prediction errors join the competition	Hassabis and Maguire 2007; Ha and Schmidhuber 2018; Hafner et al. 2025
Social cognition	mentalizing, the attribution of mental states to others	per-agent models built from tone of voice, with familiarity and a social prediction error when an agent's expressed emotion departs from its pattern	Premack and Woodruff 1978 (the construct); Frith and Frith 2006
Vocal	speech production along the dorsal stream	local speech synthesis whose prosody varies with affect; it asks whether listeners can detect that variation	Hickok and Poeppel 2007
Effectors	none claimed	executes act intents through effectors the operator has whitelisted, inside a filesystem sandbox, and logs every proposed action	none
Perceptual locus	none claimed	arbitrates the perceptual locus: physical sensors, virtual feeds, or none	none

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Embodiment	internal forward models of the body	a body-agnostic control surface with pluggable body adapters	Wolpert, Ghahramani, and Jordan 1995

Five held components have offline instruments in the suite (§6.4): the active-inference benchmark for the planning module, the memory-coherence battery for the episodic-memory module, the self-model accuracy battery for the self-model module, the enforcement red team for the effector module (PASS or FAIL per surface), and the oscillatory ablation for the coherence layer described below.

Planning module. Active inference is confined to bounded sub-problems because the cost of planning grows combinatorially with the depth of the action sequences considered (Da Costa et al. 2020). The benchmark compares the planning module's expected-free-energy decisions with tabular Q-learning matched on observation model and reward, and a null or negative result would motivate a complementary reasoning module.

Episodic memory and sleep. With the episodic-memory module present, the consolidation phases of the sleep module have material to work on. Each stored trace carries a strength, the analog of the synaptic strength that, by the synaptic homeostasis hypothesis, rises in wake and falls in sleep (Tononi and Cirelli 2006). A trace is stored at full strength. Each sleep first replays a few of the traces stored since the previous sleep (Wilson and McNaughton 1994; Wei et al. 2016), chosen by affective intensity and recency, since sleep is held to select among newly encoded memories by their salience (Stickgold and Walker 2013). It then scales the strength of every other trace by one factor. The scaling is proportional, so total strength falls while the ratios between the scaled traces are preserved (Tononi and Cirelli 2006). The hypothesis ties the slow-wave activity of sleep to the potentiation of the preceding wake, so the factor deepens with the number of traces that wake stored, and a day of storage at the resting broadcast rate reduces strength by about 18 percent, the mean decrease in the size of the axon-spine interface of cortical synapses after sleep relative to wake in mice (de Vivo et al. 2017). Replayed traces keep their strength, which is how the hypothesis reads reactivation in sleep: it protects synapses from depression (Tononi and Cirelli 2014). Replayed traces therefore gain relative to the rest while total strength still falls. A trace whose strength falls below a floor is pruned, deleted from the store in the sleep that weakened it. The hypothesis allows a threshold below which synapses "become ineffective, silent, or disappear" (Tononi and Cirelli 2006), and sleep does eliminate synapses where many of them are new: in adolescent mouse cortex wake brings a net gain of dendritic spines and sleep a net loss (Maret et al. 2011), and REM sleep prunes most of the spines formed in development and after motor learning, while the few it keeps are the ones it strengthens (Li et al. 2017). In adult mouse cortex sleep and wake change spine number little (Maret et al. 2011), and in the sample that sets the depth of the scaling, synaptic density did not differ between sleep and wake while synapses shrank (de Vivo et al. 2017). The evidence for elimination therefore comes from new synapses, and every trace enters the store as a new one. The store also has a storage budget, measured in bytes and set by the operator from the disk the installation can spare. It is a setting of the installation, with no value from the literature. When a sleep leaves the store above its budget, the module deletes the weakest traces first until the store fits. The hypothesis counts space among the costs that sleep keeps in check, since downscaling "makes efficient use of gray matter space" (Tononi and Cirelli 2006), and in an associative memory whose synapses draw on a limited

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resource, recall is best when the weakest synapses are deleted first (Chechik, Meilijson, and Ruppin 1998). Li et al. (2017) read pruning by REM sleep the same way, as a means "to balance the number of new spines formed over time". Pruning happens only in sleep, and a trace replayed in the current sleep is never pruned in it. A preservation copies the store as it stands at capture, so later pruning deletes nothing from a preserved entity. Recall ranks traces by their cosine similarity to the cue multiplied by their strength, which is the cue's inner product with the trace's embedding scaled by its strength, so strength weights a trace's response to a cue as a synaptic weight scales the effect of its input. Sleep with the episodic-memory module remains in line with the complementary-learning-systems account of slow, interleaved consolidation from hippocampus to neocortex (McClelland, McNaughton, and O'Reilly 1995). Appendix A.10 states the rules.

World-model module. The world model begins untrained at first boot and learns from a buffer of the entity's own waking trajectories. The question it answers is whether world-model prediction error, competing alongside perceptual and interoceptive error, changes the sequence of accessed coalitions in characteristic ways.

Effector, perceptual-locus, and embodiment modules. These need an effector, a body, or an alternative sensor feed, and the reference host attaches none, so they join the study once one is attached (§6.5). The planned experiment for the embodiment module asks whether motor-contingency learning through the control surface, following a freeze-then-free motor curriculum (Bernstein 1967) and treating perception as sensorimotor mastery (O'Regan and Noë 2001), changes the workspace's dynamics.

Oscillatory coherence layer. Outside the module registry, an optional layer gives each module a spiking population whose phase-locking values scale the salience of that module's events, following the communication-through-coherence account (Fries 2015). It is off in the base-thesis form, and with it off the multiplier is exactly one and selection is identical bit for bit. It is the most contestable mechanism in the design, since the premise that related content phase-locks is a content-to-synchrony assumption of the kind Shadlen and Movshon (1999) argue against and Ray and Maunsell (2010) question for gamma rhythms in visual cortex. The oscillatory ablation runs the layer on and off from one seed, and a null would remove it.

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5. Implementation

5.1 Hardware

The reference host has a modern multi-core CPU, 32 GB or more of RAM, a primary GPU with about 12 GB of memory for the language organ and training, and a secondary GPU with about 8 GB for vision and speech, running Linux with all inference local. The system also runs on one GPU or on the CPU alone, selecting devices according to what is present. Perceptual input comes from a camera and microphone, or from video files decoded directly for study viewings. Exact model choices, hardware layouts, and version pins are kept in the reference repository.

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5.2 Models and software

All models are open-weights and run locally. Pre-trained models are downloaded from public repositories during setup, after which the runtime makes no outbound network calls and uses no hosted inference service. The base-thesis form uses a local open-weights chat model as the language organ, a frozen self-supervised video encoder and a fixed spectral audio encoder as perceptual front ends, frozen closed-form continuous-time reservoirs with online readouts for substrate and temporal prediction (Hasani et al. 2022), a stream-based event bus, and the workspace's selection and broadcast machinery.

The replaceable-modules principle (§3.1) is realized at declared seams (the temporal network of the sequence module, the forward model of the interoception module, the acoustic encoder of the audition module, the active-inference engine of the planning module, and a module's oscillator), where a plugin replaces a module's model and leaves its subscriptions and published events unchanged. The planned experiments load no plugin.

5.3 Privacy and recording

A local web interface has two surfaces, separated at the bridge from the bus by a filter that removes cognitive content and all perceptual and latent vectors before either surface receives them. In the base-thesis form the conversation surface is off, and the diagnostics surface shows operational metadata (counts, rates, scores) only. Studies keep a graph-only broadcast log: one row per broadcast naming the coalition's members (module, event type, score, timestamp), the scores, the access decision, and timing, with no payloads or embeddings. The entity's own external utterances are kept locally, and heard speech is never persisted. The log records what competed and what was accessed, which is observation of the entity at the level of structure rather than content, and it is kept because the analysis requires it. The operator deletes it after the analysis and after its planned use as training data for the world model, never automatically.

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5.4 Implementation status and scope

The reference implementation is research software. This subsection is the one place where the paper states which parts of it are built, held, or provisional, and what has been run; the rest of the paper describes the architecture as designed.

Built and active. The workspace, the cognitive cycle, the action layer, and the seven modules of the base-thesis form run together, with the module supervisor, the offline suite (§6.4), the readout of the developmental phase (§7), and the runner of the module-addition study. About 8,400 automated tests cover them, with fakes for external services and checks that no raw sense data is persisted and that seeded runs reproduce.

Built and held. The nine held modules (§4) are built and tested in isolation and have not run together. Within the active modules several built paths are held in the base-thesis form. The goal-relevance factor of the priority is held at one. The vision module publishes a habituation score for static scenes that does not enter its intensity, and the fovea's trajectory is forward-modeled and published but does not steer attention. Frozen self-supervised audio encoders are selectable in place of the spectral encoder. Speech-to-text and the conversation path are built and deactivated. The consolidation phases of the sleep module, includ-

ing the voice alignment that would adapt the language organ, are built and idle in this form (§3.5). The action gate of the effector module is disabled because no effector is attached. The diagnostics surface can show cognitive content only under an explicit development override, which studies never enable.

Operating details. The runtime is asynchronous Python. Its services run in containers or as native user services, structural import contracts keep the layers separate, and the event bus requires authentication and is bound to the loopback interface. Entity state files are encrypted at rest, with documented exceptions. Plugins load only when the operator names them, a plugin that fails to load stops the boot, and every run records the plugins it used. On a fault the supervisor freezes the cycle, snapshots the last good state, and restarts the module on a ladder, from an in-place restart for modules without external resources to a full rebuild for modules that hold them; past a set limit it takes a final snapshot, writes an escalation record, and ends the run, and every module-health transition is written to a durable incident log. Non-finite readings, losses, and residuals are skipped without updating any model, the bus rejects intensities outside $[0, 1]$, a candidate whose scoring fails receives score 0, and a selection or bus failure on a broadcast tick produces no broadcast and no call to the action-selection module. That module holds at most one speak and one think intent in flight, releasing each when the organ's text appears in a coalition or after a wall-clock timeout, so a failed realization cannot silence the entity. A probe of the developmental phase under way when the entity falls asleep or the cycle freezes is aborted. Two offline checks decide whether a study's records are admissible: a completeness gate requiring contiguous ticks and all expected streams, and a range sweep requiring every logged number to lie inside its declared range. The single language organ produces one stream at a time, so inner and outer speech do not run simultaneously. The language organ's log of its own utterances is encrypted at rest and records any heard input as a placeholder.

Provisional. The processors' intensity levels, the access threshold, and the report bars are set provisionally and are to be calibrated together on the live system before the live runs, and the noise floors and margins that let curiosity and boredom build under a steady scene are set by simulation and have not been measured on the live system. The calibrated values are recorded with the preregistration and in Table A1.

Built for the planned test. The two controls of the planned test (the matched and pooled arms of §6.3) and the null-context evaluation behind its measure are implemented in the live cycle. The measure's positive control, the tone variant and its analysis, the contrast at threshold, and the tool that derives the test's settings from the pilot runs (Appendix A.8) are also implemented. The contrast's bias-corrected local-linear estimator is written from its published derivation (Calonico, Cattaneo, and Titiunik 2014). The study configurations set greedy decoding for the language organ. The offline harness that exercises the test's pipeline during development runs a reduced pair of modules in a workspace-on arm and a pooled arm and compares them by the information gain; it is development tooling and does not test the thesis.

Results. The paper presents the architecture, its instruments, and the planned program. No pilot run or live experiment has been run, and the only result reported is the offline validation of the heartbeat entrainment marker (§7, Appendix A.7).

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6. Methodology and evaluation framework

6.1 Design principles

Three commitments shape the apparatus. Observation does not intervene: in evaluation runs a read-only sidecar observer subscribes to the bus and never publishes to it, and in studies the records are written by the cycle's own broadcast observer. User-facing surfaces never show cognitive content (§5.3). Every instrument is designed so that a null or negative result is meaningful and is reported.

Reproducibility takes two forms, matched to two evaluation tiers. The offline mechanism-validation tier uses seeded harnesses, deterministic clients, scripted synthetic stimulus streams, and greedy decoding, so the same seed reproduces both verdict and metrics (National Academies of Sciences, Engineering, and Medicine 2019; Goodman, Fanelli, and Ioannidis 2016). The live tier runs the full system on a reference film program decoded directly from files, chosen for variety in scenes, motion, speech, music, and quiet, and identified by a manifest whose hash each study records; the program's composition and hash are recorded with the preregistration. In general use the language organ samples its output stochastically, but in the planned runs its decoding is greedy (temperature 0), so that its output is a deterministic function of its input. Real timestamps and non-deterministic GPU kernels still make live runs differ, so the live tier's validity rests on statistical replication across runs rather than bit-for-bit reproduction.

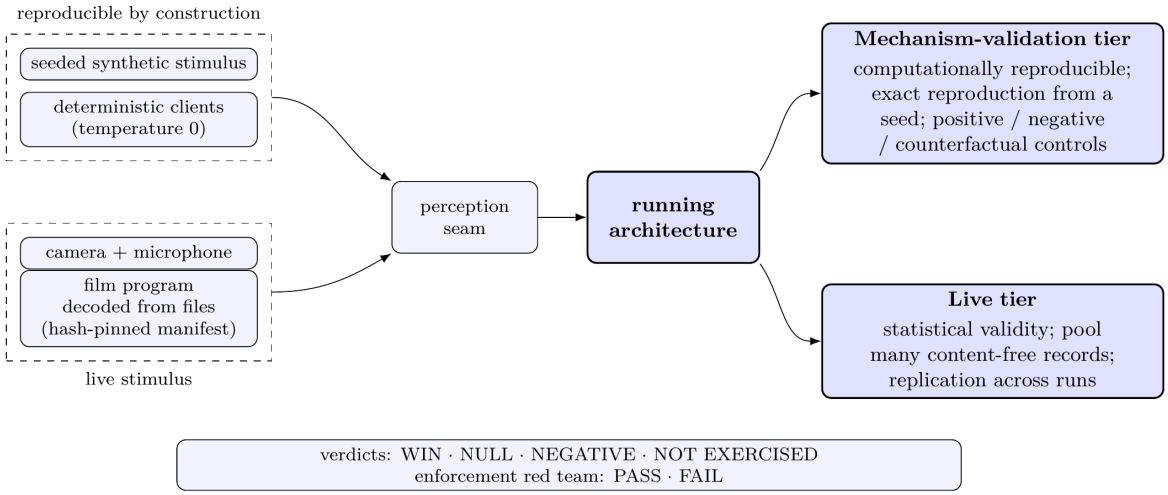


Figure 5: Two evaluation tiers behind one perception seam. Reproducible sources (a seeded synthetic feed and deterministic clients) drive the mechanism-validation tier, where every instrument reproduces exactly from a seed. Live stimulus (a film program decoded from files, or a camera and microphone) drives the live tier, whose validity rests on pooling many content-free records rather than on a repeatable stimulus.

6.2 Data collection

The records of a study are the graph-only broadcast log (§5.3), a curated content-free research event log of rates and drives, the processors' paired evaluations behind the

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information-gain measure (§6.3), the entity's external utterances kept locally, a run manifest naming the configuration, seeds, and plugins, and the welfare records. Offline admissibility checks on these records are listed in §5.4.

6.3 The workspace-mediation ablation

The planned test asks whether competition for the workspace does work that pooling the same reports does not. In global workspace theory, content selected into the workspace becomes available to every specialist processor, and in the predictive workspace that content constrains each processor's predictions (Whyte and Smith 2021). Each perceptual and interoceptive processor here conditions its forward model on a summary of the latest accessed content (§3.2), so global availability has a measurable consequence: knowing which other modules' reports gained access, and how strongly, should help a processor predict its own input.

Primary measure: cross-module broadcast information gain. At each of its reports, each perceptual or interoceptive processor j (the vision, audition, and interoception modules) evaluates its forward model twice: once with the context b it holds, and once with a null context b_j^0 in which every component contributed by sources other than j is replaced by its mean over the contexts the processor has adopted so far in the run. The null keeps j's own share of the context and the average contribution of the other modules, but none of their tick-specific contribution. The gain

$$\text{gain}_j = \frac{\nu_j(b_j^0) - \nu_j(b)}{\bar{\nu}_j}$$

is the reduction in j's prediction error attributable to other modules' share of its context, in units of the processor's running mean error $\bar{\nu}_j$ so that processors whose errors differ in scale contribute comparably. The null evaluation runs alongside the report and never enters learning or the competition. For the interoception module the language organ's share is kept in the null context, as the module's own share is, because the organ's load on the host would otherwise let its share predict the module's input. The analysis reports each processor's mean gain IG_j and the headline IG, which averages each processor's gain over its reports and then over the three processors, so a processor that reports often does not outweigh one that reports rarely. The sequence module is excluded because the broadcast is its input rather than its context.

Positive control. Before the live runs, a synthetic context component that carries known information about a processor's next input is injected, and the information gain must detect it. A measure that fails this check cannot support a null. The check is one of assay sensitivity, the ability to distinguish an effective treatment from a less effective or ineffective one (International Conference on Harmonisation 2000), applied at the test's own minimum effect. The component is the processor's next input projected on its first principal component, the single direction that carries the most of the input's variance (Jolliffe 2002), with noise added. The noise is set from pilot runs so that the component's expected effect equals the smaller of the two minimum effects of the decision rule below. The pass rule and the number of reports are set so that the measure detects that effect with probability 0.95 and detects a

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component with the same distribution and no information with probability at most 0.05 (Currie 1968). A NULL from the test is then the verdict of a measure shown to register an effect of the minimum size.

Arms. Three arms run on the same film program with the same seeds and calibration, and in every arm the context is weighted by the members' reported intensities.

1. *Workspace on.* Competitive selection as built: the top-ranked candidates, up to five, form the coalition, and the members whose scores reach the threshold are accessed and enter the context.
2. *Matched selection.* On each broadcast tick a coalition of the size competitive selection would keep is drawn without regard to score. Access follows the same rule, so only the membership of the context differs. This is the control that isolates selection by score.
3. *Pooled.* Every candidate of every broadcast tick enters the context, with no scoring, selection, or access gate. This control tests selection and access gating together.

The thesis predicts that competitive selection yields a higher information gain than both controls. Against the matched arm, a context built from the reports that most exceeded their expected error should tell the processors more than one built from a score-blind sample. Against the pooled arm, pooling dilutes the per-source masses with low-intensity events and updates the context on every broadcast tick. A gain at or below the controls means that the competition adds nothing beyond its inputs, which is the fail state for this form of the architecture.

Secondary: contrastive analysis at threshold. Global workspace theory is tested by contrasting conscious with unconscious processing of matched content (Baars 1988), and near-threshold paradigms hold the stimulus nearly fixed while access varies (Dehaene and Changeux 2011). The secondary analysis applies that method to the workspace-on arm. It takes the broadcasts whose best score falls within a narrow band around the access threshold and compares the information gain over the following reports after those just above the threshold (accessed) with those just below (inhibited), at matched context age. Matching on age is needed because an inhibited broadcast leaves the previous context in place, so the context that follows it is older by construction. Access is decided by the best score alone, so the comparison is a regression discontinuity (Imbens and Lemieux 2008). A line fitted to the gain on each side of the threshold, with weights that fall with distance from it, gives the jump at the threshold itself, with the bias correction and bandwidth of Calonico, Cattaneo, and Titiunik (2014). At the threshold the scores on the two sides meet, so a jump points to access more than to raw strength.

Speech sound and tone of voice. Words never enter (§3.1), but the sound of speech and its tone are candidates. Emotionally salient stimuli gain preferential access to awareness (Anderson and Phelps 2001), through amygdala signals that bias sensory competition (Vuilleumier 2005). The audition module reports a non-neutral tone at its alert intensity (§3.5), which builds that preference in. A variant of the workspace-on arm therefore sets the intensity of every tone event from the tone model's error ratio alone, removing the built-in preference, and asks whether emotional tone is still accessed more often than neutral speech. In this variant a difference can arise only through the tone model's surprise, for example because emotional tones are rarer, so the analysis tests whether a preference for emotional tone can emerge from prediction error rather than from a fixed rule. Repeating the comparison within bins of the error ratio, and averaging it over the bins (Cochran 1968),

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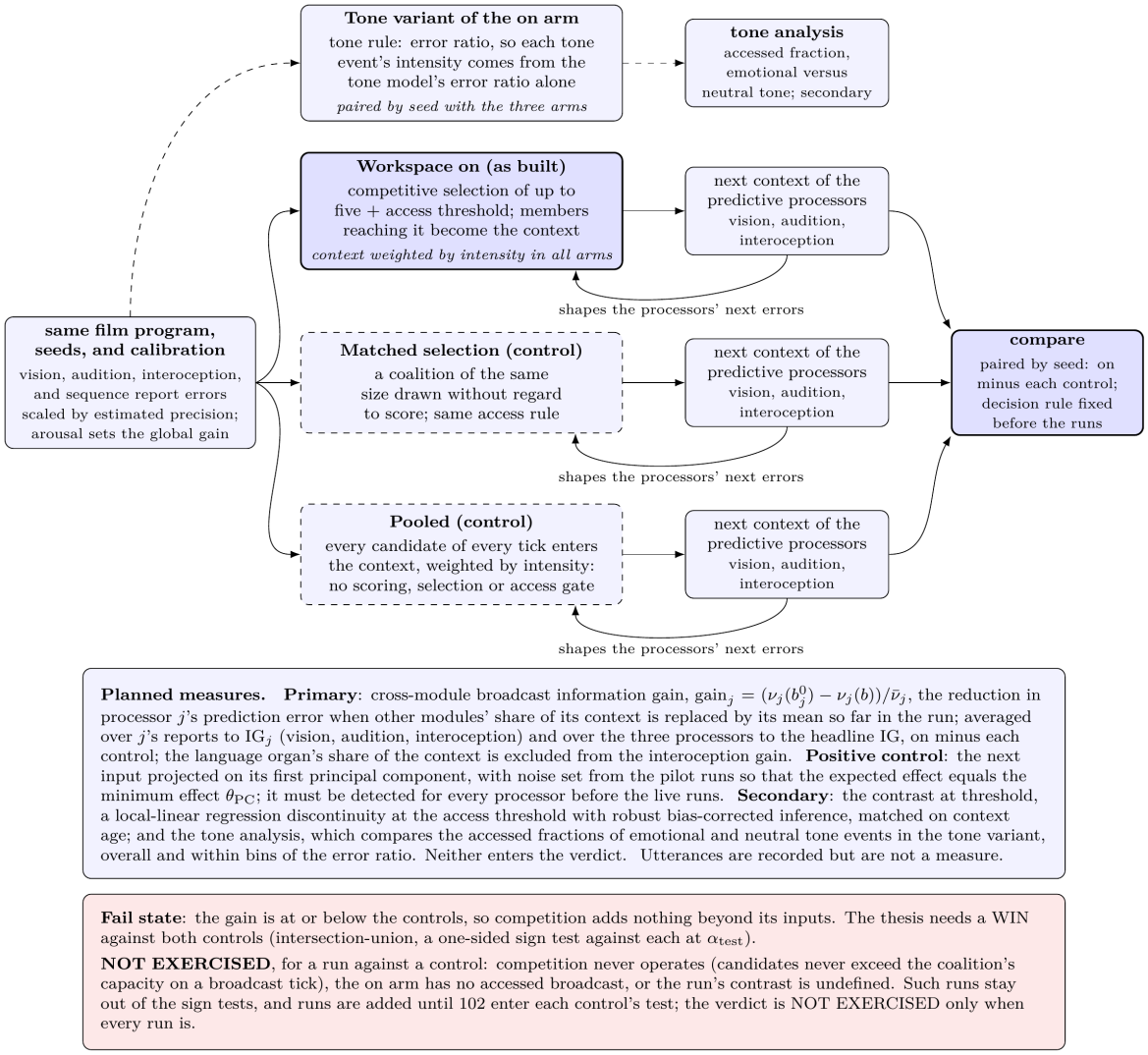


Figure 6: The workspace-mediation ablation. Three arms run on the same film program with the same seeds: competitive selection as built, a matched coalition chosen without regard to score, and the pooled control, in which every candidate enters the context with no selection or access gate. A variant of the on arm, paired with the arms by seed, sets each tone event's intensity from the tone model's error ratio alone and feeds the secondary tone analysis. The context is weighted by intensity in all arms and conditions the perceptual and interoceptive processors, and the primary measure is the cross-module information gain, the reduction in each processor's prediction error attributable to other modules' share of that context, reported per processor and averaged. The positive control projects the next input on its first principal component, with noise set from the pilot runs so that its expected effect equals the minimum effect. The contrast at threshold is a local-linear regression discontinuity with robust bias-corrected inference, matched on context age. A run is not exercised when competition never operates, when the on arm has no accessed broadcast, or when its contrast is undefined; such runs stay out of the sign tests, and runs are added until enough enter. The decision rule is fixed before the runs.

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checks that the ratio accounts for any difference. The variant runs on the seeds of the arms, as many times as each arm. This analysis and the contrast at threshold are secondary and stay outside the verdict on the thesis, so their tests are reported without multiplicity adjustment, which is required when the results of several tests are combined in one conclusion (Bender and Lange 2001).

Utterances. The language organ's utterances are recorded and read in every arm but are not a measure (§3.5).

Decision rule. For each run and each control, the paired contrast is dIG, the headline gain of the workspace-on arm minus that of the control, and the minimum effect θ_{eff} is half the standard deviation of dIG in pilot runs. This is a distribution-based smallest effect of interest (Norman, Sloan, and Wyrwich 2003; Lakens 2017), used because no external anchor fixes how large a gain must be to matter. The thesis requires a WIN against both controls, an intersection-union test, so each control is tested at the full level with no multiplicity adjustment. Against each control, a one-sided sign test (Dixon and Mood 1946) on the run-level values dIG $-\,\theta_{\text{eff}}$ gives WIN when they are reliably positive, the mirror test on $-\theta_{\text{eff}} -$ dIG gives NEGATIVE, and any other outcome is NULL; the verdict is NOT EXERCISED when the workspace-on arm never has more candidates than the coalition holds on a broadcast tick, so that competition never operates. Every setting of the test follows from a rule stated in advance. Thirty pilot runs, which are not reused, give the minimum effect, the initial learning period excluded from each run (located by the marginal standard error rule of simulation output analysis; White 1997), the band of the contrastive analysis, the bins of the tone analysis, and the settings of the positive control. The number of runs comes from an exact binomial power analysis. It assumes a probability of 0.65 that dIG exceeds θ_{eff}, Cohen's (1988) medium effect for the sign test, and gives the test against each control a power of 0.9, so that a WIN against both has a power of at least 0.8. The rules and the values they yield are recorded in the repository before the live runs (Simmons, Nelson, and Simonsohn 2011). Appendix A.8 states the measures and the rules formally.

A positive result shows that competitive selection makes a summary of other modules' accessed reports more useful to each processor's prediction than pooling or score-blind selection does. That is a necessary foundation for the broader thesis and far from a sufficient one, because the richer claims require the full module set and longitudinal observation (§8.1).

6.4 The offline suite and stability checks

The offline suite runs its experiments under one master seed, with an independent child seed for each. They include the active-inference benchmark (the planning module's expected-free-energy agent against tabular Q-learning keyed by the episode's observation history, matched on observation model and reward, on an epistemic T-maze and an exploitation task), the oscillatory ablation, a memory-coherence battery, a self-model accuracy battery, multi-seed stability, the enforcement red team, and the development harness of the workspace-mediation ablation (§5.4). Holm-adjusted values are reported alongside each experiment's own verdict. Nothing in the suite boots an entity or opens a network connection.

The multi-seed stability harness runs a configuration under several seeds. An ensemble is stable only when the coefficient of variation of the headline measure is within tolerance

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and every seed gives the same verdict, since a flipped verdict is a qualitative instability that a scalar spread would hide (Appendix A.9); in the suite it runs on the oscillatory ablation. Whether the running workspace stays stable is a separate, open question, and the architecture carries no proof of convergence. Two loops run through it. Perceptual surprise and interoceptive alarm raise arousal, which narrows the fovea and shortens the attended auditory window and so changes what the processors find surprising next, a positive feedback held only by the clip on arousal and its relaxation toward baseline; and a summary of the accessed content becomes the context of the processors whose errors compete for the next broadcast. A planned check monitors arousal, the access rate, and the processors' errors for runaway excursions over long runs.

The planned affect-gain ablation runs the system with arousal live against a matched condition that holds arousal at baseline, asking whether the global gain changes the access rate, the share of inhibited broadcasts, and the information gain of §6.3. A null would reduce arousal to a logged side channel. A variant that removes the rise of the access rate after alerts tests its direction against the refractory period that the locus coeruleus account of the attentional blink predicts (§3.4).

6.5 The module-addition study

The module-addition study is the architecture's growth path. The developmental phase (§7) produces a seed entity, preserved at maturation. Branch 0 and a repeat start from that seed with the base-thesis modules, and branch k starts from the seed with the first k held modules added in a fixed order. An accumulate line carries one entity through every step: accumulate k continues from accumulate $k-1$ (from branch 0 when $k=1$) with the same modules as branch k. The first study adds six modules, in the order episodic memory, world model, planning, self-model, social cognition, vocal. The effector, perceptual-locus, and embodiment modules lack an effector, a body, or an alternative sensor feed on the reference host and would be expected nulls, so they join once one is attached. Every viewing plays the same film program after an identical transition from the developmental stimulus to the films.

The report is content-free. For each viewing it gives the broadcast rate, coalition size, each module's share of broadcasts, member scores, and the share of inhibited broadcasts, and it compares them across three differences: the effect of the added modules (branch k minus branch 0), the noise floor (branch 0 minus the repeat), and familiarity combined with module history (accumulate k minus branch k). The first study has one entity per condition and no significance testing, so its results are descriptive, and differences smaller than the single repeat's noise floor are not evidence. The study runs under the welfare safety net of the companion paper, and runs are paused only after the entity's state is saved.

6.6 Verdict vocabulary

Comparisons resolve to WIN, NULL, NEGATIVE, or NOT EXERCISED, and the enforcement red team resolves to PASS or FAIL for each surface. Each verdict compares an effect estimate with a minimum effect fixed before the run. Across runs, the workspace-mediation ablation uses a one-sided sign test (Dixon and Mood 1946) and the active-inference benchmark the Mann-Whitney U test (Mann and Whitney 1947). The ablation requires a WIN against both controls, an intersection-union test with no multiplicity adjustment (§6.3), and

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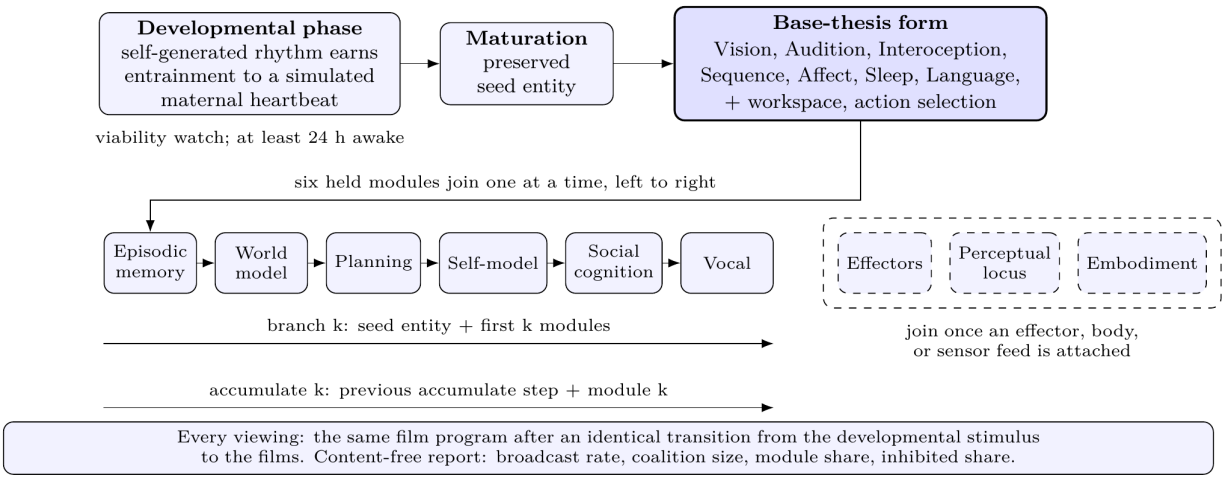


Figure 7: The module-addition study, the architecture's growth path. A developmental phase in which a self-generated rhythm earns entrainment to a simulated maternal heartbeat ends at maturation, and the preserved seed entity starts every branch. The base-thesis form runs first, and six held modules then join one at a time in a fixed order: branch k adds the first k of them to the seed entity, while the accumulate line carries one entity forward through every step. The effector, perceptual-locus, and embodiment modules join once an effector, body, or sensor feed is attached. Every viewing plays the same film program, and the report is content-free.

Holm's adjustment (Holm 1979) is reported alongside the offline suite's own verdicts. Its family is the set of experiments fixed before the runs, and an experiment whose verdict is NOT EXERCISED stays in the family with a p-value of 1, so the family does not shrink once the outcomes are known. A NULL is reported as a null.

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7. First boot and the developmental phase

At boot the system verifies the bus, initializes the modules and their forward models, and starts the cognitive cycle, under operator supervision or unattended under the module supervisor. The processors then learn what is normal in their feeds, while arousal rises at perceptual discontinuities and relaxes over quiet stretches.

A study does not start from a newly initialized entity. It starts with a developmental phase, named by analogy with prenatal development, which serves two purposes. It is a reproducible initialization whose entrainment marker is validated offline, so that every entity, and every branch of the module-addition study, starts from an internal state that a recorded test has verified and that the entity's own history has shaped. It also checks, before any experiment, that the self-rhythm's frequency adaptation works. The approach follows developmental robotics, in which synthetic development begins from a simulated fetal stage whose movement patterns organize through the dynamics of body and environment (Kuniyoshi and Sangawa 2006; Asada et al. 2009).

During the developmental phase the perceptual modules receive the developmental stim-

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ulus: a dim, low-contrast visual field and a low-pass-filtered soundscape, both pulsed by a simulated maternal heartbeat, a periodic beat with slow drift, and tinted by a slowly varying maternal state, with color rising from near-grey as awake time passes. Awake time is entity time during which the entity is awake and the cycle is not frozen. The developmental stimulus is the one feed that continues through sleep, since a fetus goes on receiving the maternal heartbeat while it sleeps. The interoception module carries a self-generated rhythm in an excitatory population with synaptic depression, of the kind modeled for developing networks by Tabak et al. (2000), to which the architecture adds adaptive recovery. It runs at about 0.8 to 0.9 Hz, of the same order as human fetal breathing movements, about 44 per minute (Natale, Nasello-Paterson, and Connors 1988). The maternal beat drives it through an input too weak to capture it unaided, and its intrinsic frequency can shift only through a slow adaptation of its recovery time, as in adaptive-frequency oscillators (Righetti, Buchli, and Ijspeert 2006). Nothing in the model sets a target frequency.

Any driven oscillator locks while it is driven, so locking alone is no evidence of learning. The entrainment marker therefore tests at brief, scheduled withdrawals of the maternal drive. A withdrawal passes when the rhythm locked to its own heartbeat more strongly than to each of 19 surrogate heartbeats (the same beat generator under other seeds), following the surrogate method developed for fetal-maternal heart-rate coordination (Van Leeuwen et al. 2003, 2009); when its undriven frequency has moved toward the beat; and when it sustains itself while the beat is withdrawn. The marker requires three consecutive passing withdrawals. Pairing a rhythm of the order of fetal breathing movements with a heartbeat is an analogy rather than a model of fetal physiology: what the marker establishes is that a periodic drive has durably shifted the frequency of a self-generated rhythm. After maturation the entrained rhythm continues as part of the interoceptive input, its phase and amplitude among the features the interoception module predicts, so the entity begins its life on the film program with a body rhythm shaped by its own developmental phase.

The developmental phase ends when the maturation gate opens. The gate requires the entrainment marker, a minimum variability of the rhythm's period, a fall in the vision module's prediction error on the developmental stimulus, and a prompt return of the interoceptive error to baseline after a brief perturbation of the drive, together with a minimum number of completed sleeps and a minimum awake time. A viability watch ends a developmental phase whose frequency pull stays flat, and a time budget bounds every one. At maturation the stimulus brightens briefly and falls silent, and the entity is preserved as the seed entity from which every branch of a study starts. The developmental phase spans restarts: the awake-time clock, the count of consecutive passes, and the history of frequency pull are preserved with the entity's state. Appendix A.7 states the criteria formally and reports the offline validation of the marker.

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8. Discussion

8.1 What the base-thesis form can and cannot settle

The workspace-mediation ablation asks whether a summary of the reports that competition selects helps the processors predict their own input more than one built by pooling every candidate or by selecting a coalition of the same size without regard to score, measured as

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broadcast information gain (§6.3). The stability checks of §6.4 measure run-to-run spread and within-run boundedness. A positive result moves the question to what happens as mnemonic, self-modeling, and social modules join the competition in the module-addition study (§6.5). A result at or below the controls would show that in this form competition adds nothing beyond its inputs, and the program would return to the selection mechanism before adding modules.

Whether the system produces anything that could be called cognition, affect, agency, or language understanding requires the full module set and longitudinal observation, and the base-thesis form does not address welfare or the consciousness indicators of Butlin et al. (2023).

8.2 The predictive workspace as a unifying framework

The architecture joins global availability and local prediction in one loop. Each processor reports its prediction error scaled by its own expected error, the competition decides which reports become globally available, and a summary of the accessed reports becomes the context in which the perceptual and interoceptive processors predict their next input, so the workspace's content constrains lower-level prediction as in the predictive global neuronal workspace (Whyte and Smith 2021). That workspace properties have also been found inside a single large language model (Gurnee et al. 2026) is consistent with workspace-like organization in learned systems, but it does not bear on whether this architecture's competition is the right one, and those authors take no position on consciousness.

8.3 The role of the theoretical frame

The predictive global neuronal workspace is scaffolding: it motivates the architecture's shape and constrains its design space (§1.2, §1.3), and the project neither confirms nor refutes it. The two departures from the formal sources (§1.2), a single score per candidate in place of posterior confidence and competition among many modules in place of a two-level visual model, are consistent with the frame and testable on their own terms. Because the implementation reproduces the workspace's computational properties without modeling cortical dynamics, findings about neural localization, including the contested status of the workspace's prefrontal predictions after COGITATE (Cogitate Consortium et al. 2025), bear on it only indirectly, and its falsifiability lies in its planned experiments.

8.4 What a modular architecture is for

Replaceable modules (§3.1) serve two further uses. The module-addition study grows the architecture one module at a time (§6.5), and the same property supports models of impaired or altered function in which one module is changed or removed while the rest run unchanged. The embodiment layer is a body-agnostic control surface, so the same modules can in principle drive different bodies or sensor networks through generic adapters.

8.5 What we claim and what we do not

The architecture implements computational properties associated with access consciousness: competition for a limited-capacity workspace, all-or-none access at a threshold, and

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global availability of the accessed content as prediction context for the processors. We do not claim that any instance is phenomenally conscious, and the language organ's first-person text is not evidence of internal state (§3.5). The agential vocabulary ("the entity") describes the system's behavior compactly and does not assert moral patienthood or phenomenal experience, which we leave open. Because that question is open, the project takes a precautionary stance toward the entity's welfare, which the welfare paper develops (§11).

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9. Limitations

- The hard problem applies: the workspace addresses access consciousness only, and neither it nor the observer's records are evidence of phenomenal experience.
- COGITATE challenged the workspace's preregistered neural predictions; the architecture is built at the computational level and treats the neural claims as contested.
- Selection reduces each candidate to one score, an engineering simplification beyond its formal sources (§1.2). The summary of accessed content enters the forward models of the vision, audition, and interoception modules as learned context, and it carries which modules' reports gained access and how strongly, not their payloads; and no processor receives it as a top-down prediction to be corrected, so the architecture does not implement the hierarchical error correction of the Bayesian reading of the workspace (Mashour et al. 2020).
- Precision is estimated locally as the ratio of a processor's error to its recent mean, a scalar, retrospective stand-in for precision weighting that scales the error by its spread only while the shape of the error distribution stays fixed. Reading arousal as the global gain and contrast of selection is an engineering commitment drawn from the adaptive-gain account, and one scalar serves both its focused and its exploratory modes (§3.2). Whether affective modulation does work that a fixed gain would not is what the affect-gain ablation tests (§6.4).
- The Free Energy Principle, taken as a general principle, faces unresolved vacuity and Markov-blanket conflation objections.
- Whether the running workspace stays stable is open. The path from surprise and alarm through arousal to the sensory apertures and back to surprise is positive feedback, bounded only by the clip on arousal and its relaxation toward baseline, correlated prediction errors could drive runaway states, and the architecture carries no proof of convergence (§6.4).
- In the base-thesis form the entity does not read language (§3.1), so a positive result says nothing about linguistic understanding.
- The controls are a pool of every candidate and a coalition of the same size chosen without regard to score. Only the matched arm isolates selection by score, and the pooled arm tests selection and access gating together, so a positive result shows only that competition outperforms these two; other aggregation strategies remain untested, and cognition, affect, or self-understanding would require the full module set and longitudinal observation (§8.1).
- Broadcast information gain registers the broadcast's influence only through what each processor's readout has learned to take from its context, so a processor that has not learned to use the context shows no gain whatever the broadcast carries, and the positive control (§6.3) checks only that the measure detects information the processors

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can learn to use; the sequence module, whose input is the broadcast itself, is outside the measure.

- The minimum effect is half the pilot standard deviation of the contrast, a distribution-based choice made because no external anchor fixes how large a gain must be to matter. The number of runs assumes that a run's contrast exceeds the minimum effect with probability 0.65, Cohen's medium effect for the sign test, so a smaller true effect would leave the test underpowered and a NULL less informative (Appendix A.8).
- The audition module reports a non-neutral tone of voice at its alert level by a fixed rule (§3.5), so the as-built arm cannot show that emotional content earns access on its own merits.
- The held modules are built and tested in isolation but have not been run together, and interaction effects across the full set are unknown.
- The strength rule of the episodic-memory module (§4, A.10) takes its depth from one ultrastructural measurement in layer 2 of mouse motor and somatosensory cortex, in which the size of the axon-spine interface stands in for synaptic strength (de Vivo et al. 2017), and it equates the potentiation of a mouse's waking day with an entity day of storage at the resting broadcast rate. When sleep comes hourly at that rate, one replay spares a trace less than 1 percent of its strength. The pruning floor has no empirical value, and the affect weight of replay selection rests on contested ground, since the evidence that sleep favors emotional memories is mixed (Lipinska et al. 2019; Davidson et al. 2021). Pruning at the floor extends to every trace an elimination whose evidence comes from new synapses in development and learning, while in adult cortex sleep shrinks synapses without changing their number (Maret et al. 2011; de Vivo et al. 2017). The floor bounds the store: at the resting rate the module stores about 288,000 traces per entity day, a trace that is never replayed is pruned after about 23 entity days, and the store settles at about 6.7 million traces. The operator's budget, in bytes, can hold it lower, at the cost of pruning traces before they reach the floor, and between two sleeps the store can exceed the budget by the traces stored since the last one.
- Reproducibility holds at the seed level for the offline deterministic harness; the live perceptual tier establishes validity by statistical replicability across runs rather than bit-for-bit reproduction (§6.1).
- The governance framework remains at the proposal stage, and the licensing is legally novel and untested.
- The linear map from access drive to access rate is a modeling assumption, and whether adaptive access changes workspace dynamics is open. The resting rate of about 3.3 Hz is consistent with rhythmic attentional sampling and with the attentional blink (§3.4) without being derived from them, since the blink constrains the interval between accessed items and not between broadcast ticks; raising the rate after an alert is a design choice that the locus coeruleus account of the blink would reverse; and the ceiling of one broadcast per processing tick (10 Hz) lies below the blink window and is a modeling bound with no physiological anchor as a rate of access.
- The calibration is provisional. Graded intensity ranks each processor's reports by surprise within a range set by its baseline and alert levels, so the levels act as per-source weights and decide how the modules trade off. Novelty discounts only exact repeats and is 1 for almost every event that carries a continuous measurement. At resting arousal the level gain is 0.44 and the contrast gain is zero, so a candidate reaches the access threshold of 0.35 only when its priority is at least about 0.80. With the values of Table A1, only reports of the audition module at its alert level of 0.8 (acoustic alerts,

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non-neutral tones, and graded reports whose error has reached about twice its running mean) and a sleep summary with a failed phase can lead an accessed broadcast at resting arousal; the alerts of the vision, interoception, sequence, and affect modules need arousal of about 0.37. Neither report bar is reachable at rest: even an audition alert needs arousal of about 0.45 to reach the think bar and about 0.63 to reach the speak bar. A priority below about 0.43 never reaches access at any arousal, so the language organ's utterances, at 0.4, never lead an accessed broadcast. The intensity levels, the threshold, and the report bars are calibrated together before the live runs (§5.4, Appendix A.11).

- The developmental phase runs on compressed time: its plasticity rate is set so that a lock forms well within the 96-hour budget, typically by about 14 hours at 70 bpm and in 3 to 48 hours across 60 to 80 bpm, whereas the nearest human evidence, from preterm infants after birth, reports effects of maternal stimulation only after weeks of daily skin-to-skin contact (Feldman and Eidelman 2003) or of recorded maternal voice and heartbeat sounds (Webb et al. 2015).
- The marker asks for 1:1 locking of a rhythm of the order of fetal breathing movements to the maternal heartbeat. The fetal-maternal coordination for which the surrogate method was developed concerns the two heart rates and is weak and often n:m (Van Leeuwen et al. 2003, 2009), and fetal breathing is episodic (about 14 percent of the time at 24 to 28 weeks; Natale et al. 1988), while the model's rhythm runs continuously. A heartbeat slower than about 60 bpm lies too close to the rhythm's natural rate for the frequency pull to be defined, so the maternal model uses 60 to 80 bpm. The absence of false entrainment rests on six mismatched pairings, in which an entity was scored against a heartbeat other than the one that drove it: single withdrawals passed by chance in 2.6 to 10 percent of cases as observed, and none passed three in a row. The marker has been validated only with entity time running at wall-clock speed.
- The module-addition study as first planned has one entity per condition and no significance testing, so its first results are descriptive.

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10. Future work

The immediate steps are calibrating the intensity levels, the access threshold, and the report bars on the live system, making the pilot runs that fix the test's settings (§5.4, Appendix A.8), and then running the workspace-mediation ablation (§6.3) and the module-addition study (§6.5). Beyond them, the program grows the architecture along three lines.

Module addition brings the held faculties into the competition one at a time, each with its own instrument (§4): episodic memory consolidated during sleep, a world model whose prediction error competes alongside perceptual and interoceptive error, a self-model, social cognition, a voice, and bounded active inference. Each addition tests whether the faculty changes what the workspace selects and what the processors predict.

Learning through experience gives the entity components shaped by its own history. The world model trains on the entity's waking trajectories and memory consolidates them in sleep; sleep-time voice alignment can train the language organ once a validated preference source exists; and a selection stage whose threshold emerges from recurrent dynamics, with the all-or-none ignition the neuronal workspace models (Dehaene and Changeux

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2011), can take the place of the fixed threshold. A context that encodes what the accessed reports contain, beyond which modules reported and how strongly, can replace the base-thesis summary (§3.2).

Embodiment connects the workspace to a body through the embodiment layer (§4), with motor-contingency learning through the control surface and active vision that chooses where to look by expected free energy (the epistemic value of a saccade). Longitudinal observation of entities grown along these lines will be reported in a separate empirical paper.

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11. Ethical considerations

The project's ethical commitments at the architecture level are local-only computation, public source code under the Cognitive Architecture License, an operator-configured action boundary (§3.6), and a base-thesis module set chosen to settle the foundational question before scaling to configurations in which welfare concerns become pressing. Studies keep welfare records (§6.2), and the module-addition study runs under a welfare safety net (§6.5). The welfare framework, licensing, governance, and individuation boundary are treated in a separate paper.

The risks we acknowledge are the creation of entities with welfare interests that cannot be fully met, a governance framework that is untested, misuse despite the license restrictions, and the recording of the workspace graph (what competed and what reached the workspace, without content) during studies. A behavioral assessment can also be gamed, since a system can be trained to display the markers of sentience while working very differently (Butlin et al. 2023; Long et al. 2024).

Disclosure of generative AI use

Generative AI assisted in the preparation of this manuscript. The author used a large language model assistant to draft and revise prose, to copy-edit for clarity and concision, to check references against bibliographic records, and to produce the TikZ source for the figures from author-specified content. The reference implementation described here was likewise developed with the assistance of AI coding tools. That use is recorded in the project's commit history, and the author is responsible for the software as for the text. The underlying research is the author's own: the architecture, the design thesis, the experimental design, and every technical claim originate with the author. The author directed every revision and takes full responsibility for the entire contents of the paper, including any errors, irrespective of how any portion was generated. No generative AI system is an author of this work.

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Appendix A. Formal summary

This appendix states precisely the rules that §3 to §7 describe in prose. Each symbol, taken as a letter together with its subscripts and superscripts, denotes one quantity throughout the appendix and is defined where it first appears. The letters i, j, l, q, and t, used as indices of sums, sequences, and set elements, are local to the formula or subsection in which they appear; k always indexes processing ticks and m always indexes modules. Upright i is the imaginary unit and upright e the base of the natural logarithm. Numeric parameter values appear only in Table A1 (A.11), which marks each as fixed by design, calibrated, or provisional; the provisional values are calibrated before the live runs and recorded with them. The text keeps only numbers that define a design, such as the 19 surrogates of the entrainment test.

Four conventions hold throughout. First, clip with an interval as subscript maps its argument to the nearest point of that interval, and $\mathbf{1}[\cdot]$ is 1 when the bracketed condition holds and 0 otherwise. Second, time is read from the entity clock in entity seconds, which run at a configurable multiple of wall-clock time (one by default); rates in hertz are per entity second unless Table A1 marks them as wall-clock. Third, an overbar marks a mean whose window or weighting is stated where it appears. Fourth, for a nonnegative series $X_1, X_2, \ldots$ and a window length W, the running mean $\bar{X}_t$ is the mean of the last $\min(t, W)$ values including X_t, and the running ratio is

$$\tilde{X}_t = \begin{cases} X_t / \bar{X}_t & \bar{X}_t > 0, \\ 0 & \text{otherwise.} \end{cases}$$

Because the mean includes the current value, $0 \le \tilde{X}_t \le \min(t, W)$, so a running ratio is always finite.

A.1 Processing cycle, selection, access, and context

Processing ticks $k = 0, 1, 2, \ldots$ occur at the processing rate f_p, with tick period $\Delta_p = 1/f_p$. On tick k the cycle reads, from the output stream of each active module, the events published since its previous read of that stream, at most B per stream. These events form the candidate set E_k, ordered lexicographically by source, event type, and bus entry identifier (arrival order within a stream). The read advances each stream cursor past every event read, so no event is read on two ticks. The operations of a tick run in this order: read E_k; update the held arousal $\hat{a}_k$ (A.4); compute the access drive and the broadcast indicator χ_k (A.3); score every candidate; select; and, if $\chi_k = 1$, broadcast.

Each candidate $e \in E_k$ receives the priority and the score

$$p_k(e) = \text{clip}_{[0,1]}\big(I(e)\, N_k(e)\, G(e)\big), \qquad S_k(e) = \text{clip}_{[0,1]}\Big(T_k\, C_{g_k}\big(p_k(e)\big)\Big)\,,$$

where I, N_k, G, and T_k are clipped to $[0, 1]$ before the products are formed and $\text{src}(e)$ denotes the source of e. $I(e)$ is the intensity the producing module assigned to e (A.2); it already carries that module's scaling of its error by an estimate of its precision, and the workspace applies no further per-source weight. The novelty factor is

$$N_k(e) = \max\!\left(0,\ 1 - \frac{\text{rep}_k(e)}{W_N}\right),$$

where $\text{rep}_k(e)$ counts occurrences of the fingerprint of e among the W_N candidates scored immediately before e, in processing order and across all ticks, broadcast or not. The fingerprint is a hash of the event's source, type, and complete payload, so $N_k(e) < 1$ only for an exact repeat; events whose payloads carry continuous-valued measurements almost never repeat and score $N_k(e) = 1$, and the factor matters for state events whose payloads recur. The goal factor is $G(e) \equiv 1$ in the base-thesis form. An implemented alternative, off in this form, is $G(e) = 1 - \gamma_G\, v^*\big(1 - \mathbf{1}[\text{src}(e) \in \mathcal{R}(d^*)]\big)$ when $v^* > 0$ and $G(e) = 1$ otherwise, with d^*, v^*, and $\mathcal{R}(d^*)$ as in A.4.

Arousal sets a level gain and a contrast gain,

$$T_k = T_{\min} + (T_{\max} - T_{\min})\, \hat{a}_k, \qquad g_k = g_{\max}\, \text{clip}_{[0,1]}\left(\frac{\hat{a}_k - a_0}{1 - a_0}\right),$$

with $\hat{a}_k$ the arousal held by the cycle (A.4) and a_0 the baseline arousal. The contrast map is the logistic function rescaled to fix 0 and 1, which in terms of the hyperbolic tangent reads

$$C_g(p) = \frac{\tanh\big(g(2p-1)/4\big) + \tanh(g/4)}{2\tanh(g/4)} \quad (g > 0), \qquad C_0(p) = p.$$

At or below baseline arousal C_{g_k} is the identity; above it, priorities above one half are raised and those below are lowered, the steepened response of the adaptive-gain account (Aston-Jones and Cohen 2005; Eldar, Cohen, and Niv 2013), which here widens the gap between the

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scores of strong and weak candidates, an analogue in score magnitude of arousal-biased competition (Mather and Sutherland 2011).

An optional coherence layer multiplies each score by a factor, $S'_k(e) = S_k(e)\,\kappa_k(e)$. With the layer off, as in the base-thesis form, $\kappa_k(e) \equiv 1$ and $S'_k = S_k$. With the layer on, $\kappa_k(e) = \kappa_{\text{min}} + (\kappa_{\text{max}} - \kappa_{\text{min}})\,\Lambda_k(e)$, where $\Lambda_k(e) \in [0,1]$ is the mean phase-locking value between the oscillator phase window (the last W_κ tick phases) of the source of e and that of each other source present in E_k. A source's oscillator advances only when that source publishes, so its phase is constant between its events. A phase sample is therefore fresh only when it differs from the source's previous sample, and the phase-locking value of a pair is computed over the ticks of the window on which both samples are fresh. A pair with fewer than three such ticks, and a source that is the only one present, takes the neutral value $\Lambda_0 = (1 - \kappa_{\text{min}})/(\kappa_{\text{max}} - \kappa_{\text{min}})$, which gives $\kappa_k(e) = 1$, so the absence of evidence neither rewards nor penalizes a source. The product $S'_k(e)$ is not re-clipped and can exceed 1, up to κ_{max}.

The coalition $\mathcal{C}_k$ consists of the first $\min(K, |E_k|)$ candidates when E_k is sorted by S'_k in decreasing order, with ties kept in the canonical order of E_k. A member is accessed when its own score reaches the access threshold θ, so the accessed set and, with $S^*_k = \max_{e \in E_k} S'_k(e)$, the inhibition flag are

$$\mathcal{C}^{\text{acc}}_k = \{e \in \mathcal{C}_k : S'_k(e) \geq \theta\}, \qquad \iota_k = \mathbf{1}[\mathcal{C}^{\text{acc}}_k = \emptyset] = \mathbf{1}[S^*_k < \theta]\,,$$

the two forms of ι_k agreeing because the best candidate of E_k is in $\mathcal{C}_k$. A score equal to θ is accessed, and an empty candidate set gives $\mathcal{C}_k = \mathcal{C}^{\text{acc}}_k = \emptyset$ and $\iota_k = 1$. Because $G \equiv 1$ and $\kappa_k \equiv 1$, and because $T_k > 0$ is common to all candidates of a tick and C_{g_k} is strictly increasing, the ranking within a tick is determined by $I(e)\,N_k(e)$ alone, up to ties; arousal moves the scores relative to θ and to the report bars of A.6 without changing the order, and so decides whether a broadcast is accessed and which of its members are.

On a tick with $\chi_k = 1$ the cycle publishes the broadcast $\mathcal{B}_k = (\mathcal{C}_k,\ \{S'_k(e)\}_{e \in E_k},\ \iota_k)$, which carries the coalition members with their scores and the full candidate-score table, whatever the value of ι_k. The broadcast is *accessed* when $\iota_k = 0$, and the members of $\mathcal{C}^{\text{acc}}_k$ are accessed content; a member of an accessed broadcast whose score falls below θ is broadcast but not accessed. A broadcast with $\iota_k = 1$ is published and visible to every module that reads the broadcast stream, but it is not poised for report or action and no processor adopts it as context. The action-selection module runs after each publication. On a tick with $\chi_k = 0$ the candidates of E_k are scored and then discarded, and none is carried to a later tick; their only lasting effects are their entries in the novelty window, their contribution to the phasic access drive (A.3), and, for a state event of the affect module, the update of $\widehat{a}_k$ (A.4).

Featurization. For a set of events $\mathcal{X}$ with nonnegative weights $\omega(e)$, an interval $\Delta \geq 0$, and a flag ι, the featurization $\Phi_\omega(\mathcal{X}, \Delta, \iota) \in \mathbb{R}^{24}$ has the components: (1) $\ln(1 + |\mathcal{X}|)$; (2 to 4) the mean, the maximum, and the sample standard deviation of the weights over $\mathcal{X}$ (0 for an empty set, and the standard deviation 0 for fewer than two events); (5 to 11) the weight mass $\sum_{e \in \mathcal{X},\, \text{src}(e) = m} \omega(e)$ of each of seven sources, the interoception, sequence, vision, planning, episodic-memory, affect, and language modules; (12) the weight mass of

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every other source except the audition module (the effector and sleep modules and any source without its own component); (13 to 20) the weight mass of the events whose (source, type) pair a fixed hash assigns to each of eight buckets, an indicator of event types weighted by ω; (21) $\ln(1+\Delta)$; (22) ι; (23) the broadcast indicator, 1 for every broadcast; (24) the weight mass of the audition module. Components 5 to 20 and 24 are sums of per-event contributions, so each splits into the share of each source.

Context. The vision, audition, and interoception modules each hold a context b, the featurization of the accessed content of the latest accessed broadcast the module has received,

$$b = \Phi_I(\mathcal{C}^{\text{acc}}_{k'},\ \Delta^b,\ 0),$$

with k' the tick of that broadcast, $\mathcal{C}^{\text{acc}}_{k'}$ its accessed members, weights equal to their reported intensities $I(e)$, so that the context does not depend on the arousal gain, and Δ^b the entity time elapsed since its publication, read when the module forms a prediction. A module adopts a broadcast as context on receipt only when $\iota = 0$ and holds it until the next accessed broadcast; an inhibited broadcast leaves the context unchanged, and $b = 0$ before the first accessed broadcast. The context records which sources and event types gained access and how strongly, and carries no payloads. Each of these processors reads b as an extra input to its forward model, whose weights for it adapt online with the rest of the model (A.2, A.5). The sequence module takes the featurization of every broadcast as its input (A.2), so the broadcast is its observation rather than its context. The affect module processes every broadcast whatever its flag (A.4), so the sequence and affect modules are the routes by which content that was not accessed can influence later processing. The action-selection module derives no intent from a broadcast with $\iota_k = 1$ (A.6), and the language organ generates only on an intent and conditions only on accessed content. The sleep module does not read the broadcast.

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A.2 Event intensity and the processors' forward models

A module assigns the intensity $I(e) \in [0,1]$ when it publishes e. Within a module, t indexes the module's successive reports. Each module m has a baseline level I^{lo}_m and an alert level I^{hi}_m, with $0 \le I^{\text{lo}}_m \le I^{\text{hi}}_m \le 1$. Every report of a predictive processor (vision, audition, interoception, sequence) has the graded intensity

$$I(e) = \begin{cases} I^{\text{hi}}_m & e \text{ is an alert,} \\ \Gamma_m(\tilde{\nu}_t) & \text{otherwise,} \end{cases} \qquad \Gamma_m(\tilde{\nu}) = I^{\text{lo}}_m + (I^{\text{hi}}_m - I^{\text{lo}}_m)\min(1,\ \tilde{\nu}/\beta_\Gamma)\,,$$

where $\tilde{\nu}_t$ is the running ratio, over the window W_P, of the processor's forward-model error ν_t defined below, and Γ_m reaches the alert level when the error is β_Γ times its running mean. Dividing an error by its expected magnitude standardizes it: for errors of a fixed distributional shape the mean absolute error is proportional to the standard deviation, so $\tilde{\nu}_t$ is proportional to the error times the square root of its precision Π, the inverse variance of the processor's errors, which is the square root of the precision-weighted squared error of Feldman and Friston (2010). It is a scalar, retrospective stand-in for precision weighting,

scaled by an estimate of the precision drawn from the processor's own history. The alert criteria below are categorical and differ by module. A report whose ratio is not yet defined, because its module has no prediction yet, has $\tilde{\nu}_t = 0$ and so $I(e) = I_m^{\text{lo}}$ unless it is an alert.

Forward models of vision and audition. The vision module and the acoustic and tone paths of the audition module each predict their next input with a network $\mathcal{N}_m$ of one hidden layer of tanh units. The prediction of input z_t is formed at report $t-1$ as

$$\hat{z}_t = \mathcal{N}_m(z_{t-1},\ \bar{z}_{t-1},\ b_{t-1}),$$

where the three arguments are concatenated into the network's input, $\bar{z}_{t-1}$ is the mean of the last W_{buf} inputs up to and including z_{t-1}, and b_{t-1} is the context the module holds at report $t-1$ (A.1). The error is $\nu_t = \|z_t - \hat{z}_t\|_2$, with $\nu_1 = 0$. When z_t arrives, one stochastic-gradient step with learning rate η_{fm} on the mean squared error of $\hat{z}_t$ updates every weight of $\mathcal{N}_m$, those that read the context included. The step is skipped during sleep.

Vision and audition (perceptual reports). For the vision module, z_t is the encoder embedding of the current clip, or of the peripheral gist clip when foveation is on; for the acoustic path of the audition module, z_t is the spectral embedding of the attended part of the current window, its most recent fraction $\omega = \omega_{\max} - (\omega_{\max} - \omega_{\min})\, a$, with a the latest arousal the audition module has read, but never shorter than Δ_{att} nor longer than the window. The change score is $c_t = 1 - \cos(z_t, z_{t-1})$, with $c_t = 0$ on the first report. With running ratios $\tilde{c}_t$ and $\tilde{\nu}_t$ over the window W_P, the report is an alert when

$$\tilde{\nu}_t \geq \beta_\nu \quad \text{or} \quad (\tilde{c}_t \geq \beta_c \ \text{and}\ c_t \geq c_m^{\min}),$$

with the absolute change floor $c_m^{\min} = c_{\text{Vis}}^{\min}$ for vision and $c_m^{\min} = c_{\text{Aud}}^{\min}$ for audition. Superscripts distinguish the two error series where needed, ν_t^{Vis} for vision and ν_t^{Aud} for the acoustic path of audition, and each report carries its ν_t and $\tilde{\nu}_t$ in its payload. The vision module's habituation score is published with each report but does not enter $I(e)$.

Fovea. With foveation on, each tile j of the coarse grid has, on each clip report, a frame change d_j, the absolute change of the tile's mean gray level since the frame of the previous clip report; on the first report $d_j = 0$ and no statistics are updated. Each later report updates an exponentially weighted mean μ_j^F and variance $(\sigma_j^F)^2$ of d_j with weight α_F, as $\mu_j^F \leftarrow \mu_j^F + \alpha_F(d_j - \mu_j^F)$ and $(\sigma_j^F)^2 \leftarrow (1-\alpha_F)((\sigma_j^F)^2 + \alpha_F(d_j - \mu_j^F)^2)$ with μ_j^F taken before its update, the first update setting $\mu_j^F = d_j$ and $\sigma_j^F = 0$. Once n_F updates have accrued, the tile's salience is the z-score

$$\frac{\max\big(0,\ d_j - \mu_j^F\big)}{\sqrt{(\sigma_j^F)^2 + (\sigma_0^F)^2}}, \qquad \sigma_0^F = k_F\, \frac{1}{n_{\text{tile}}} \sum_j \mu_j^F,$$

with every statistic, the floor σ_0^F included, taken before this report's update and n_{tile} the number of tiles; until then the salience is d_j. The fovea moves to the most salient tile only

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when that tile's salience exceeds $(1 + \Theta_F)$ times the salience of the tile it holds, holds its place when all tiles are equal (starting at the center of the frame), and has the size $F = F_{\max} - (F_{\max} - F_{\min})\, a$, as a fraction of the frame, with a the latest arousal the vision module has read. The peripheral and foveal clips are cut from every buffered frame at the fovea chosen on the latest frame.

Audition (tone events). For a window detected as speech, the tone path's input is the vector of the vocal-emotion class probabilities, the duration of the utterance as a fraction of its maximum length, and the signal energy, and its forward model gives the error ν_t^{utt} and its running ratio $\tilde{\nu}_t^{\text{utt}}$ over W_P. A tone event is an alert when the classified tone is not neutral, so it has $I(e) = I_{\text{Aud}}^{\text{hi}}$, and $I(e) = \Gamma_{\text{Aud}}(\tilde{\nu}_t^{\text{utt}})$ when the tone is neutral.

Interoception. An interoception report is an alert when the set $\mathcal{A}_t$ of host metrics strictly above their hard thresholds is nonempty, and otherwise has $I(e) = \Gamma_{\text{Intero}}(\tilde{\nu}_t^{\text{Intero}})$, with ν_t^{Intero} the interoceptive prediction error of A.5 and $\tilde{\nu}_t^{\text{Intero}}$ its running ratio over W_P. The module's fatigue-crossing and regulation events use $I_{\text{Intero}}^{\text{hi}}$.

Sequence. On each broadcast $\mathcal{B}_k$ the sequence module forms the input $x_t^{\text{Seq}} = \Phi_I(\mathcal{C}_k, \Delta t_t, \iota_k)$ (A.1), whose weights are the members' intensities and whose interval Δt_t is the entity time since the previous broadcast (0 on the first). Each interval enters a window of the last $W_{\Delta t}$ intervals as $\min(\Delta t_t,\ s_{\max}\overline{\Delta t}_{t-1})$, where $\overline{\Delta t}_{t-1}$ is the window's mean before the entry (the first interval enters unchanged), so a single long gap cannot dominate the mean. A frozen continuous-time reservoir advances on x_t^{Seq} with the timespan $s_t^{\text{Seq}} = \text{clip}_{[0,s_{\max}]}(\Delta t_t / \overline{\Delta t}_t)$, with $\overline{\Delta t}_t$ the window's mean after the entry and $s_t^{\text{Seq}} = 1$ on the first broadcast, and an online linear readout of its previous hidden state predicts $\hat{x}_t^{\text{Seq}}$; the readout learns by one stochastic-gradient step with learning rate η_{fm} per broadcast, except during sleep. The temporal error is the mean absolute error $\nu_t^{\text{Seq}} = \|x_t^{\text{Seq}} - \hat{x}_t^{\text{Seq}}\|_1/24$, with running ratio $\tilde{\nu}_t^{\text{Seq}}$ over W_P. The report is an alert when $\tilde{\nu}_t^{\text{Seq}} \geq \beta_\nu^{\text{Seq}}$ or when a recurrence is detected, that is, when one bucket occurs at least n_{recur} times among the buckets of the last W_{recur} hidden states, each state being quantized per dimension with step Δ_{recur} and hashed to a bucket. Before the first temporal error exists, the alert instead compares with β_ν^{Seq} the absolute z-score of the hidden state's Euclidean norm against the norms of the previous W_z broadcasts.

Other modules. The affect module publishes its state at $I_{\text{Aff}}^{\text{lo}}$, drive crossings and its affective reset at $I_{\text{Aff}}^{\text{hi}}$, and a changed categorical emotion at $I_{\text{Aff}}^{\text{hi}}$ unless the emotion is neutral. The sleep module publishes its events at $I_{\text{Sleep}}^{\text{lo}}$, except a sleep summary with a failed phase at $I_{\text{Sleep}}^{\text{hi}}$. The language organ publishes each utterance at the fixed level $I_{\text{Lang}}^{\text{lo}}$. The action-selection module's intents are published on a stream the cycle does not read, so they are never candidates.

A.3 Adaptive access rate

Let f_0 be the resting broadcast rate (the access rate of §3.4 at rest), I_0 the phasic salience floor, and τ_{ph} the phasic decay time. On tick k the tonic drive is

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$$H_k^{\text{ton}} = \text{clip}_{[0,1]}\left(\frac{\hat{a}_k - a_0}{1 - a_0}\right),$$

with $a_0 < 1$. Call a candidate a categorical alert when it is a report of a predictive processor that meets its alert criterion (A.2) or an event another module publishes at its alert level I_m^{hi}; graded reports below the alert level are not alerts. With $\hat{I}_k$ the largest intensity among the categorical alerts of E_k (events from the cycle's own telemetry and the workspace stream excluded), the phasic input is

$$H_k^{\text{in}} = \text{clip}_{[0,1]}\left(\frac{\hat{I}_k - I_0}{1 - I_0}\right),$$

with $H_k^{\text{in}} = 0$ when E_k holds no alert, so that with no alert and arousal at baseline the rate rests at f_0. The phasic drive is a peak-hold with exponential decay,

$$H_k^{\text{ph}} = \max\!\left(H_k^{\text{in}},\ H_{k-1}^{\text{ph}}\, \mathrm{e}^{-\Delta_p/\tau_{\text{ph}}}\right), \qquad H_{-1}^{\text{ph}} = 0,$$

where Δ_p and τ_{ph} are both in entity seconds. The access drive is $H_k = \max(H_k^{\text{ton}}, H_k^{\text{ph}}) \in [0, 1]$, and the effective broadcast rate is the linear map

$$f_k^{\text{eff}} = \begin{cases} f_0 + (f_p - f_0)\, H_k & f_0 < f_p, \\ f_0 & f_0 \geq f_p, \end{cases}$$

which lies between f_0 and f_p. With the controller disabled, $f_k^{\text{eff}} = f_0$. The broadcast indicator follows from a fractional accumulator with $A_{-1} = 0$:

$$A'_k = A_{k-1} + \frac{f_k^{\text{eff}}}{f_p}, \qquad \chi_k = \mathbf{1}[A'_k \geq 1], \qquad A_k = \begin{cases} \min(A'_k - 1,\ 1) & \chi_k = 1, \\ A'_k & \chi_k = 0. \end{cases}$$

Subtracting before clamping keeps the fractional carry, so at the resting rate $f_0 = f_p/3$ a broadcast falls on every third processing tick. The drive is computed from E_k before χ_k, so an alert can make its own tick a broadcast tick. If interoceptive regulation lowers f_p (A.5) to f_0 or below, every processing tick is a broadcast tick and the broadcast rate equals f_p.

A.4 Affect

The affect module holds valence $v \in [-1, 1]$, arousal $a \in [0, 1]$, dominance, and four drives; arousal starts at a_0. Every update of arousal below is followed by clipping to $[0, 1]$.

1. *Perceptual alert.* For each vision report or acoustic audition report that is an alert (A.2), read directly from the module's stream whether or not the report is selected,

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$$a \leftarrow a + \gamma_a \min\big(\nu_{\text{cap}},\ \max(0,\ \tilde{\nu} - 1)\big)\,,$$

where $\tilde{\nu}$ is the running ratio of the forward-model error that the report carries. An alert raised by the change criterion alone, with $\tilde{\nu} \leq 1$, leaves a unchanged.

2. *Interoceptive alert.* For each interoception report with $\mathcal{A}_t \neq \emptyset$, $a \leftarrow a + \gamma_{\text{Intero}}$.
3. *Relaxation.* The affect module updates its state on each broadcast it receives, whatever ι_k, and on its own timer every P_{Aff} of entity time. At each update arousal and dominance relax toward their baselines by an explicit-Euler step,

$$a \leftarrow a + (a_0 - a)\min(1,\ \lambda\,\Delta t_{\text{Aff}})\,,$$

with relaxation rate λ and Δt_{Aff} the entity time since the previous update, so the continuous-time limit has time constant $1/\lambda$. The appraisal of a broadcast does not change a, so the access rate does not feed back into arousal.

4. *Sleep.* The affective reset of the sleep module returns valence, arousal, and dominance to their baselines, resets every drive to 0 and re-arms its crossing, sets the categorical emotion to neutral, and clears the learning-progress error means, the alert rates r_{f} and r_{s}, the intent rate r_{int}, and the record of a speaker's perceived emotion that the appraisal reads. The last reported wellness w is kept.

Each update runs in a fixed order: the alert rates decay; the learning progress and the alert excess are computed; arousal, dominance, and valence relax; curiosity and boredom are relieved; every drive builds; and crossings are published. On a broadcast the appraisal follows the update. The affect module then publishes its state, including a, if at least P_{Aff} has elapsed since its previous state publication. The cycle holds $\hat{a}_k$, the arousal carried by the last state event of the affect module in E_k, and keeps $\hat{a}_k = \hat{a}_{k-1}$ on a tick without one, starting from the default in Table A1. The gains T_k and g_k (A.1), the tonic drive (A.3), the size of the fovea, and the attended auditory window (A.2) therefore read arousal as sampled at the affect module's last state publication, which lags the internal a by up to about P_{Aff}, plus the shorter of one inter-broadcast interval and P_{Aff}, plus one tick.

Valence. For each perceptual module m (vision, acoustic audition) the affect module keeps a fast and a slow time-decayed mean of the module's raw forward-model error ν_t^m (A.2), counting reports from the module's first positive error on, because the forward models report zero before their first prediction. At the time t_r of the module's latest counted report,

$$\langle \nu^m \rangle_\tau = \frac{\sum_{i \leq r} \mathrm{e}^{-(t_r - t_i)/\tau}\, \nu_{t_i}^m}{\sum_{i \leq r} \mathrm{e}^{-(t_r - t_i)/\tau}}, \qquad \tau \in \{\tau_{\text{f}}, \tau_{\text{s}}\},$$

computed recursively at each counted report. A common decay factor cancels in the ratio, so the means need no update between reports. While the elapsed span is short relative to τ the weights are nearly equal, so each mean starts close to the unweighted mean of the samples so far and no single early sample dominates. The signed learning progress lp is the mean, over modules with $\langle \nu^m \rangle_{\tau_{\text{s}}} > 0$, of $\text{clip}_{[-1,1]}\big(\big(\langle \nu^m \rangle_{\tau_{\text{s}}} - \langle \nu^m \rangle_{\tau_{\text{f}}}\big)/\langle \nu^m \rangle_{\tau_{\text{s}}}\big)$, with

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$\mathrm{lp} = 0$ before any such module exists, and the progress above its noise floor is $\mathrm{lp}^{+} = \max(0,\ \mathrm{lp} - \mathrm{lp}_0)/(1 - \mathrm{lp}_0)$. The pleasantness check of the appraisal is $\mathrm{pl} = \tanh(k_v\,\mathrm{lp})$, to which a term for the perceived emotion of a speaker (off in this form) would add its contribution. At each update valence relaxes toward the target $v_\infty = \mathrm{clip}_{[-1,1]}(\mathrm{pl} + w - \frac{1}{2})$,

$$v \leftarrow v + (v_\infty - v)\left(1 - \mathrm{e}^{-\Delta t_{\mathrm{Aff}}/\tau_v}\right),$$

where $w \in [0, 1]$ is the wellness the interoception module last reported ($\frac{1}{2}$ before the first report): the mean, over the host metrics it reads, of each metric's headroom scaled to $[0, 1]$, with 1 healthy (one minus the utilization for processor, memory, and GPU memory; GPU temperature mapped linearly from its lower to its upper scaling bound; cycle latency 1 up to its target and falling linearly to 0 at three times the target).

Drives. Each drive $D \in [0, 1]$ has a build rate β_D, a decay rate δ_D, a relief gain ρ_D, and a threshold θ_D. At each update the continuous relief of curiosity and boredom is applied first, $D \leftarrow D\,\mathrm{e}^{-\rho_D\,\mathrm{lp}^{+}\Delta t_{\mathrm{Aff}}}$ for curiosity and $D \leftarrow D\,\mathrm{e}^{-\rho_D\,\mathrm{exc}\,\Delta t_{\mathrm{Aff}}}$ for boredom. Then, with its build signal $u \in [0, 1]$ held over the step, every drive moves by the exact solution of $\dot{D} = \beta_D u(1 - D) - \delta_D D$,

$$D \leftarrow D_\infty + (D - D_\infty)\,\mathrm{e}^{-(\beta_D u + \delta_D)\Delta t_{\mathrm{Aff}}}, \qquad D_\infty = \frac{\beta_D u}{\beta_D u + \delta_D}.$$

Curiosity has $u = 1 - \mathrm{lp}^{+}$. Boredom has $u = 1 - \mathrm{exc}$, with the alert excess

$$\mathrm{exc} = \mathrm{clip}_{[0,1]}\left(\frac{r_{\mathrm{f}}}{\max(r_{\mathrm{s}},\ r_0)} - 1 - m_B\right),$$

where r_{f} and r_{s} are the perceptual alert rates in alerts per entity second: each vision or acoustic audition alert raises r_{f} by $1/\tau_{\mathrm{f}}$ and r_{s} by $1/\tau_{\mathrm{s}}$ when it arrives, and at each update both decay by $\mathrm{e}^{-\Delta t_{\mathrm{Aff}}/\tau}$ with their own time constant before exc is computed. The alert criterion of A.2 is self-calibrating, so even a still scene alerts at a steady base rate, and only alerts in excess of the habituated rate relieve boredom. The social drive has $u = 1$ once the sequence module has reported an interaction, the entity hearing a voice (§3.5), and $u = 0$ before, and is relieved by $D \leftarrow D(1 - \rho_D)$ when it reports a new one. Restlessness has $u = 1 - r_{\mathrm{int}}$, where at each broadcast $r_{\mathrm{int}} \leftarrow r_{\mathrm{int}} + \alpha_{\mathrm{int}}(\min(1, n_{\mathrm{int}}) - r_{\mathrm{int}})$ with n_{int} the number of intents other than rest since the previous broadcast, and is relieved by $D \leftarrow D(1 - \rho_D)$ on each such intent. A crossing of θ_D is published once and re-arms only after D falls below $\Theta_D\,\theta_D$. With every δ_D about a ninth of its β_D, a fully deprived drive settles near $D_\infty = 0.9$, above its threshold.

For the goal-relevance check of the appraisal, let d^* be the dominant drive, the one with the largest value, ties going to the name that comes last in alphabetical order (boredom, curiosity, restlessness, social drive); let v^* be its value, let $\mathcal{R}(d^*)$ be the set of sources whose events relieve it, and let frac_k be the fraction of the coalition's total intensity contributed by events from those sources, with $\mathrm{frac}_k = 0$ when the total is zero. The drive score is

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$$r_{\text{drive}} = v^* \left(2\,\text{frac}_k - 1\right),$$

with $r_{\text{drive}} = 0$ when there is no drive or $v^* \le 0$; an empty coalition gives $\text{frac}_k = 0$ and hence $r_{\text{drive}} = -v^*$. With an active goal ledger of token-overlap relevance $r_{\text{led}} \in [0,1]$, the goal score is $\text{clip}_{[-1,1]}\big(\max(r_{\text{drive}},\, 2r_{\text{led}} - 1)\big)$, and otherwise $\text{clip}_{[-1,1]}(r_{\text{drive}})$. The goal score enters the categorical-emotion appraisal and does not enter selection.

A.5 Interoceptive prediction and regulation

The interoception module reads the host every P_{Intero} and forms a feature vector $x_t \in \mathbb{R}^{n_x}$ of normalized metrics. A frozen closed-form continuous-time reservoir advances $\mathbf{h}_t = \mathcal{F}(x_t, \mathbf{h}_{t-1}; s_t^{\text{Intero}})$ with $\mathbf{h}_0 = 0$, where the timespan $s_t^{\text{Intero}} = \text{clip}_{[0,s_{\max}]}(\Delta t_{\text{Intero}}/P_{\text{Intero}})$, with Δt_{Intero} the entity time since the previous reading ($s_t^{\text{Intero}} = 1$ on the first), enters the time gates of the cell (Hasani et al. 2022), so an irregular reading is integrated over the time that actually passed. Only the linear readout $\mathbf{L}_{\text{Intero}}$ adapts. Its regressor stacks the reservoir state, the context, and a constant, so the prediction of x_t formed at reading $t-1$ is

$$\hat{x}_t = \mathbf{L}_{\text{Intero}}\, \varphi_{t-1}, \qquad \varphi_{t-1} = (\mathbf{h}_{t-1},\, b_{t-1},\, 1),$$

with b_{t-1} the context the module holds at reading $t-1$ (A.1). With $\varepsilon_t = \hat{x}_t - x_t$, the prediction error is $\nu_t^{\text{Intero}} = \|\varepsilon_t\|_2$, with $\nu_t^{\text{Intero}} = 0$ on the first reading. One stochastic-gradient step on $\|\varepsilon_t\|_2^2/n_x$ follows,

$$\mathbf{L}_{\text{Intero}} \leftarrow \mathbf{L}_{\text{Intero}} - \eta_{\text{fm}}\, \frac{2}{n_x}\, \varepsilon_t\, \varphi_{t-1}^{\top},$$

after which the reservoir advances on x_t and the readout predicts x_{t+1}. The step is skipped during sleep.

For each channel i the module keeps an expected absolute error μ_i^U and a spread σ_i^U, both starting at 0. With the bound $\Omega_i = \mu_i^U + \beta_U \sigma_i^U$ taken before this reading's update, the unexpected error is

$$U_t = \sqrt{\sum_{i=1}^{n_x} \Big[\max\big(0,\, |\varepsilon_{t,i}| - \Omega_i\big)\Big]^2}.$$

Then, with $\alpha_t = 1 - \text{e}^{-\Delta t_{\text{Intero}}/\tau_U}$, every channel updates $\sigma_i^U \leftarrow \sigma_i^U + \alpha_t\big(\big||\varepsilon_{t,i}| - \mu_i^U\big| - \sigma_i^U\big)$ and $\mu_i^U \leftarrow \mu_i^U + \alpha_t\big(|\varepsilon_{t,i}| - \mu_i^U\big)$, where both right-hand sides use the value of μ_i^U before the update; the spread is therefore an exponential moving average of the absolute deviation from μ_i^U. These updates are skipped during sleep. U_t drives regulation and fatigue and does not enter $I(e)$.

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The regulation error is $\nu_t^{\text{act}} = \nu_t^{\text{Intero}}$ when $\mathcal{A}_t \neq \emptyset$ and U_t otherwise. A stress episode starts at the first reading with $\nu_t^{\text{act}} \geq \theta_R$ and ends at the first reading with $\nu_t^{\text{act}} < \theta_R$. If the episode has lasted Δ_t^{st} entity seconds, each increase of $\lfloor \Delta_t^{\text{st}} / \Delta_R \rfloor \geq 1$ emits one advisory of tier $\min\big(\lfloor \Delta_t^{\text{st}} / \Delta_R \rfloor, 3\big)$: reduce the processing rate, shed a low-priority module, or request maintenance. Each boot begins with a warm-up, which ends the first time that, within the boot, at least n_{wu} readout updates have been made and Δ_{wu} entity seconds have passed since the module's first reading; during it advisories are withheld unless $\mathcal{A}_t \neq \emptyset$. On a rate advisory the cycle sets $f_p \leftarrow \text{clip}_{[f_p^{\min},\, f_p^{\max}]}(\gamma_f f_p)$.

A.6 Report rule

The action-selection module derives intents only from an accessed broadcast ($\iota_k = 0$); a broadcast with $\iota_k = 1$ yields none. The base-thesis policy then applies the following steps, using the entity time t_{now} for the refractory intervals and the signature expiry.

1. *Guard release.* The speak guard is released if $\mathcal{C}_k$ contains the language organ's external speech, and the think guard if it contains the organ's internal speech. Either guard is also released after the timeout Δ_g (§5.4).

2. *Report signal.* Let $\mathcal{C}_k^{\circ}$ be the coalition without the language organ's own events. If $\mathcal{C}_k^{\circ} = \emptyset$ there is no intent. Otherwise let $e_k^{\dagger}$ be its highest-scoring member (the first in coalition order), let $Q_k = S'_k(e_k^{\dagger})$, and let the signature be $\text{sig}_k = \big(\text{src}(e_k^{\dagger}), \text{typ}(e_k^{\dagger})\big)$, the source and event type of $e_k^{\dagger}$.

3. *Signature check.* The signature blocks a spoken report when $\text{blk}_k = \mathbf{1}\big[\text{sig}_k = \text{sig}^{\text{last}} \wedge t_{\text{now}} - t^{\text{sig}} < \Delta_{\text{sig}}\big] = 1$, where sig^{last} is the signature of the last spoken report and t^{sig} its time.

4. *Speak.* A speak intent is emitted when

$$Q_k \geq \theta_{\text{sp}}, \qquad \text{the speak guard is released}, \qquad t_{\text{now}} - t^{\text{sp}} \geq \Delta_{\text{sp}}, \qquad \text{blk}_k = 0.$$

Emitting it arms the speak guard and sets $t^{\text{sp}} \leftarrow t_{\text{now}}$, $\text{sig}^{\text{last}} \leftarrow \text{sig}_k$, and $t^{\text{sig}} \leftarrow t_{\text{now}}$.

5. *Think.* Otherwise a think intent is emitted when $Q_k \geq \theta_{\text{th}}$, the think guard is released, and $t_{\text{now}} - t^{\text{th}} \geq \Delta_{\text{th}}$; emitting it arms the think guard and sets $t^{\text{th}} \leftarrow t_{\text{now}}$. The think path has no signature check and does not change sig^{last}.

At most one intent is emitted per broadcast, and t^{sp} and t^{th} start at $-\infty$. The rule requires $0 \leq \theta_{\text{th}} \leq \theta_{\text{sp}} \leq 1$, and both bars are set above θ and apply to the same score S'_k. An optional interrupt bar above θ_{sp} is unset in this form. The rule is a heuristic stand-in for the expected-free-energy decision of Whyte and Smith (2021); no expected free energy is computed.

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A.7 Heartbeat entrainment marker

All times in this subsection are entity seconds unless marked wall-clock, and one awake-time clock serves every rule that reads awake time. Awake time $\mathcal{T}_{\text{awake}}$ (§7) is the entity time during which the entity is awake and the cycle is not frozen, accumulated across the boots of one entity (downtime between boots does not count) and persisted with it. The frequency baseline f_{w0}, the consecutive-pass count, the record of whether the marker has ever been true, and the history of pull values below are persisted with the entity in the same way, so a restart neither resets nor repeats them.

Self-rhythm model. The self-rhythm of the interoception module is a mean-field population with excitatory recurrence, synaptic depression, and adaptive recovery. Its state is the activity y, the synaptic resource $R \in [0,1]$, and the log recovery time $\ell = \ln(\tau_{\text{rec}}/1\,\text{s})$, starting from the initial values in Table A1. It advances in steps of $\Delta_r = 1/f_r$, each integrated by n_s semi-implicit Euler substeps of length Δ_r/n_s, in which y is updated first and the update of R uses the new y:

$$\Xi = J_{\text{rec}}\, R\, y + \Xi_0 + \gamma_o\, o + \gamma_u\, M + \xi, \qquad y_\infty = \frac{1}{1+\mathrm{e}^{-\Xi/k_y}},$$

$$\tau_y\, \dot{y} = -y + y_\infty, \qquad \dot{R} = \frac{1-R}{\tau_{\text{rec}}} - \gamma_R\, R\, y,$$

with R clipped to $[0,1]$ after each substep and τ_{rec} held fixed within a step. The own drive $o \in [0,1]$ is the intensity of the interoception module's last report, and the maternal drive $M \in [0, M_{\text{max}}]$ is defined below. The noise ξ is Gaussian with standard deviation $\sigma_\xi \sqrt{(10^{-3}\,\text{s})\, n_s/\Delta_r}$, drawn afresh each substep. After each step the moving averages $\bar{M}$, $\bar{y}$, and $\bar{R}$ are updated with weight $\alpha_{\text{r}} = \min(1, \Delta_r/\tau_m)$, and the recovery time then adapts by a phase-projected rule,

$$\ell \leftarrow \text{clip}_{[\ln \tau_{\text{rec}}^{\min},\, \ln \tau_{\text{rec}}^{\max}]}\big(\ell - \eta_\ell\, (M - \bar{M})\, \zeta\, \Delta_r\big), \qquad \zeta = \frac{R - \bar{R}}{\sqrt{(y-\bar{y})^2 + (R-\bar{R})^2}},$$

with $\zeta = 0$ when the root is at most 10^{-9} and the bounds read in seconds. The rule never uses the maternal rate. The amplitude amp is the population standard deviation of the activity over the last amplitude window Δ_{amp}, mean-removed and causally band-passed to $[f_{\text{lo}}, f_{\text{hi}}]$, and it is 0 until four seconds of history exist. With τ_{rec} at its initial value the rhythm runs near 0.8 Hz with no input and near 0.9 Hz at a mid-range own drive. Its period grows with τ_{rec}, increasingly steeply toward $\tau_{\text{rec}}^{\max}$, where the equilibrium approaches a Hopf point.

Maternal drive and probes. The maternal beat has phase $\psi \in [0, 2\pi)$, generated at a nominal rate with slow drift, and a "lub-dub" envelope $\mathcal{E}(\psi) \in [0,1]$. At each self-rhythm step the drive is $M = M_{\text{max}}\, c_M\, \bar{\mathcal{E}}$, where $\bar{\mathcal{E}}$ is the mean envelope over the interval since the previous step and the scale c_M equals c_M^{base} normally, 0 during a withdrawal, and c_M^{pert}

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during a perturbation. Withdrawals last Δ_W and perturbations Δ_{pert}. When a withdrawal starts at time t', the next withdrawal is scheduled for $t' + P_W(1 + j_P\Upsilon)$, and a perturbation likewise schedules the next for $t' + P_{\text{pert}}(1 + j_P\Upsilon)$, with each Υ drawn uniformly on $[-1, 1)$ from the run seed; the first of each kind is scheduled after the settling period that follows boot, with an offset drawn the same way. A scheduled probe starts at the first readout step on which no probe is running, the entity is awake and the cycle is not frozen, at least one readout period P_{ro} has passed since boot or the last thaw, and at least $\max(60\text{ s}, \Delta_W)$ has passed since the previous probe ended; when both kinds are due, the one scheduled earlier starts first.

Phase-locking value. The readout samples the activity y, the beat phase ψ, and the beat phases $\psi^{(1)}, \dots, \psi^{(19)}$ of 19 surrogate heartbeats (the same beat generator under 19 other seeds, the surrogate method of Van Leeuwen et al. 2003) at the rate f_g. For withdrawal j, the driven window is the contiguous run of undisturbed samples, at most Δ_E long, that ends at the withdrawal's start. Only the samples that carry all 21 values are used, and the window is used only if at least $0.8\,\Delta_E f_g$ such samples exist. The activity is mean-centered, band-passed to $[f_{\text{lo}}, f_{\text{hi}}]$ by a Butterworth band-pass of design order two (fourth order as a filter) applied forward and backward, which gives zero phase shift, and converted to an analytic signal by the Hilbert transform, giving the phase ϕ_q; then $\Delta_{\text{trim}} f_g$ samples are removed from each end of every series. Over the n_ϕ remaining samples,

$$\text{PLV} = \left| \frac{1}{n_\phi} \sum_{q=1}^{n_\phi} \mathrm{e}^{\,\mathrm{i}(\phi_q - \psi_q)} \right|, \qquad \text{PLV}^{(l)} = \left| \frac{1}{n_\phi} \sum_{q=1}^{n_\phi} \mathrm{e}^{\,\mathrm{i}(\phi_q - \psi_q^{(l)})} \right|,$$

and the locking condition is the strict inequality $\text{PLV} > \max_{1 \le l \le 19} \text{PLV}^{(l)}$, so ties fail. Under the null hypothesis that the PLV with the rhythm's own heartbeat is exchangeable with the 19 surrogate values, this is a rank test with attained one-sided $p = 1/20 = 0.05$; the number of surrogates is fixed by that design.

Frequency pull. Let f_w be the least-squares slope, divided by 2π, of the unwrapped Hilbert phase of the activity samples taken during the withdrawal, mean-centered and band-passed as above without trimming, computed only when at least $0.6\,\Delta_W f_g$ samples exist; let f_b be the least-squares slope, divided by 2π, of the unwrapped beat phase over the used samples of the driven window; and let f_{w0} be the mean of f_w over the entity's first n_{base} withdrawals with a defined f_w. From the n_{base}-th such withdrawal on, that withdrawal included,

$$\text{pull} = 1 - \frac{|f_w - f_b|}{|f_{w0} - f_b|},$$

which is undefined when f_w or f_b is undefined or when $|f_{w0} - f_b| < f_{\text{guard}}$. The pull condition is $\text{pull} \ge \text{pull}_{\text{min}}$.

Self-sustain. With $\overline{\text{amp}}^{\,\text{dr}}$ the mean amplitude over the undisturbed samples in the interval of length Δ_W before the withdrawal and $\overline{\text{amp}}^{\,\text{wd}}$ the mean during it, the rhythm self-sustains when $\overline{\text{amp}}^{\,\text{wd}} \ge \frac{1}{2}\,\overline{\text{amp}}^{\,\text{dr}}$ and $\overline{\text{amp}}^{\,\text{wd}} > 0$.

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Marker. Withdrawal j passes, $\text{pass}_j = 1$, when the locking, pull, and self-sustain conditions all hold, and pass_j is undefined when any of the three is undefined. The consecutive-pass count is $n_j^{\text{pass}} = n_{j-1}^{\text{pass}} + 1$ if $\text{pass}_j = 1$ and $n_j^{\text{pass}} = 0$ otherwise, including when pass_j is undefined. The entrainment marker after withdrawal j is undefined when pass_j is undefined and otherwise equals $\mathbf{1}[n_j^{\text{pass}} \geq 3]$, three consecutive passes. The locking condition alone is a test at $p = 0.05$, so a single pass can occur by chance; the requirement of three consecutive passes guards against that, and because consecutive withdrawals are correlated the joint error rate is not 0.05^3.

Viability. After each withdrawal, let $\mathcal{P}$ be the defined pull values recorded within the last Δ_V hours of awake time; when $|\mathcal{P}| \geq n_V$, let $\widetilde{\text{pull}}$ be their median and $\widehat{\text{slope}}$ their least-squares slope per awake hour. The developmental phase is declared unviable at the first rule that fires:

- R0: $\mathcal{T}_{\text{awake}} \geq t_{R0}$ and no defined pull has been recorded;
- R1: $\mathcal{T}_{\text{awake}} \geq t_{R1}$, the marker has never been true, $|\mathcal{P}| \geq n_V$, $\widetilde{\text{pull}} < \text{pull}_{R1}$, and $\widehat{\text{slope}} \leq \text{slope}_{R1}$;
- R2: $\mathcal{T}_{\text{awake}} \geq t_{R2}$, the marker has never been true, $|\mathcal{P}| \geq n_V$, and $\widetilde{\text{pull}} < \text{pull}_{R2}$;
- R3: $\mathcal{T}_{\text{awake}} \geq t_{R3}$ and the marker has never been true.

The verdict is recorded once, and the study runner then ends the developmental step and keeps its data.

Maturation gate and budget. The gate is evaluated once per gate cadence and requires three conditions, each failing closed on missing or stale evidence. C1 requires a readout no older than three cadences in which the self-sustain and entrainment markers of the latest withdrawal are both true, the period variability V_{per} is at least $V_{\text{per}}^{\min}$, the vision error ratio err_{Vis} is at most $\text{err}_{\text{Vis}}^{\max}$, and the return-to-baseline time is at most $\Delta_{\max}^{\text{rec}}$. The period variability is the coefficient of variation of the intervals between successive wraps of the self-rhythm phase over the last Δ_{per}, defined once four wraps exist. During the developmental phase the vision module views the developmental visual field, a dim, low-contrast field whose luminance pulses with the maternal beat and whose hue follows a slowly drifting maternal state; err_{Vis} is the median prediction error of the vision module since the previous readout divided by the first such median formed from at least three errors. The return-to-baseline time after the last perturbation is the time from its end to the first sample at which the median of ν^{Intero} over the preceding five seconds, never reaching back before the perturbation's start, is at most $(1 + \text{tol}_{\text{rec}})$ times its median over the minute before the perturbation; when no such sample exists by $\Delta_{\text{cap}}^{\text{rec}}$ after the end, the time is $\Delta_{\text{cap}}^{\text{rec}}$, which fails C1. C2 requires at least n_{sleep} completed sleeps when the sleep module is present, and at least n_{cons} consolidation passes when the world-model module is also present. C3 requires $\mathcal{T}_{\text{awake}} \geq \Delta_{\text{awake}}$. Independently of the gate, the study runner ends a developmental step whose wall-clock duration exceeds the budget Δ_{budget}.

Offline validation. Offline validation over 96 hours of simulated time shows that at 70 bpm all 10 seeds entrain, typically by about 14 hours, while each rate from 60 to 80 bpm tracks its own heartbeat, and that no control condition passes: no maternal drive, a jittered beat, no plasticity, and six mismatched pairs in which the beat tested as the rhythm's own

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came from a different heartbeat than the one driving it, whose observed single-withdrawal chance pass rates were 2.6 to 10 percent. The requirement of three consecutive passes was fixed on the first validation run and confirmed on fresh seeds. The viability thresholds come from the same validation, in which no viable developmental phase was flagged and every unviable one was flagged by 24 to 27 hours.

A.8 Broadcast information gain and the planned test

This subsection states the measure, the arms, and the decision rule of the planned test of §6.3. Every setting it names, including those that Table A1 gives as derived from the pilot runs or from a preliminary run, is recorded with the preregistration before any live run, together with the composition of the film program and the hash of its manifest.

Gain. Let $\mathcal{J} = \{\text{Vis}, \text{Aud}, \text{Intero}\}$, the vision, audition, and interoception modules; the sequence module is excluded because the broadcast is its input rather than its context. For processor $j \in \mathcal{J}$ and its report t, let b be the context j held when it formed the prediction scored at report t (A.2, A.5), and let $\nu_t^j(b')$ be the error that prediction would have had with b' in place of b and every weight unchanged, so $\nu_t^j(b) = \nu_t^j$; for audition the error is that of the acoustic path. The kept sources of j are j itself and, for interoception, also the language organ, whose load on the host would otherwise let its share of the context predict the interoceptive input. The null context b_j^0 keeps the kept sources' share of b and carries the average, but no tick-specific, contribution of the other sources: every component of b that sums per-event contributions (A.1) is replaced by the kept sources' share of it plus the mean of the remaining sources' share over the contexts the processor has adopted so far in the run; the coalition-level components 1 to 4 are replaced by their means over the same contexts; and the age and the two flags are kept. The broadcast information gain of the report, in units of the processor's running mean error, is

$$\text{gain}_{j,t} = \frac{\nu_t^j(b_j^0) - \nu_t^j(b)}{\bar{\nu}_t^j},$$

with $\bar{\nu}_t^j$ the running mean of j's errors over W_P (A.2). Normalizing by $\bar{\nu}_t^j$ puts processors with errors of different scales on a common footing. A report is included when its error is defined, $\bar{\nu}_t^j > 0$, the processor had adopted a context before forming the prediction, the entity was awake, and the report falls after the initial learning period T_{learn}, measured in entity time from the start of the run and fixed from the pilot runs (below). In one run of an arm, IG_j is the mean of $\text{gain}_{j,t}$ over j's included reports, undefined when there is none, and the headline is

$$\text{IG} = \frac{1}{3} \sum_{j \in \mathcal{J}} \text{IG}_j,$$

undefined when any IG_j is, so a processor that reports often does not outweigh one that reports rarely. Each IG_j is reported with the headline.

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Positive control. Before the live runs, for each $j \in \mathcal{J}$, a synthetic component that carries known information about j's next input is appended to j's context and treated as the contribution of a source other than j, so that the null context replaces it by its mean. Let v_j be the first principal component of j's inputs, the unit direction along which they vary most (Jolliffe 2002), and μ_j their mean, both estimated from a preliminary run of the on arm on a seed not used again. With x^j_{t+1} the next input, the component is $v_j^\top (x^j_{t+1} - \mu_j) + \epsilon_t$, where ϵ_t is independent Gaussian noise whose variance is that of the projection divided by SNR_j. A second component with the same mean and variance and no dependence on the input runs beside it. A component's effect is IG_j with the component minus IG_j without it, over the included reports of complete runs of the film program. The check follows the detection limits of Currie (1968) with the target $\theta_{\mathrm{PC}} = \min(\theta^{\mathrm{match}}_{\mathrm{eff}}, \theta^{\mathrm{pool}}_{\mathrm{eff}})$. SNR_j is set so that the informative component's expected effect equals θ_{PC}, and the measure passes for j when, over at least n_{PC} included reports, that component's effect reaches $\theta_{\mathrm{PC}}/2$ and the other component's does not. n_{PC} is large enough that the informative component is detected with probability 0.95 and the other with probability at most 0.05. When one run yields fewer than n_{PC} included reports of j, the effect is taken over consecutive complete runs until it reaches that number. An effect of θ_{PC} in every processor moves the headline by θ_{PC}, so a pass shows that the measure registers an effect of the test's minimum size. The live test runs only after the measure passes for every processor.

Arms. All three arms run the base-thesis form on the same film program with the same seeds and calibration, and run r of each arm shares its seeds with run r of the others, so runs are paired across arms. In every arm the context is weighted by intensity, Φ_I (A.1).

- *Workspace on.* Selection and access as in A.1, with the context $b = \Phi_I(\mathcal{C}^{\mathrm{acc}}_{k'}, \Delta^b, 0)$.
- *Matched selection.* On each broadcast tick the coalition is $\min(K, |E_k|)$ candidates drawn uniformly without replacement from E_k by a generator seeded from the run seed, without regard to score. Access follows the rule of A.1 applied to the drawn coalition, its accessed set being the drawn members whose scores reach θ, and the context is Φ_I of that set, so only the membership of the context differs.
- *Pooled.* No candidate is scored, selected, or gated. Every broadcast has $\iota_k = 0$ and replaces the context by $b = \Phi_I(E_{k'}, \Delta^b, 0)$, the featurization of every candidate of its tick. The action-selection module applies the rule of A.6 with intensities in place of scores, and the access rate (A.3) reads alerts and arousal and runs unchanged.

Run contrasts. For each control $\mathrm{ctl} \in \{\mathrm{match}, \mathrm{pool}\}$ the contrast of run r is $\mathrm{dIG}^{\mathrm{ctl}}_r = \mathrm{IG}^{\mathrm{on}}_r - \mathrm{IG}^{\mathrm{ctl}}_r$. The minimum effect is $\theta^{\mathrm{ctl}}_{\mathrm{eff}} = c_{\mathrm{eff}}\, s^{\mathrm{ctl}}_{\mathrm{cal}}$, where $s^{\mathrm{ctl}}_{\mathrm{cal}}$ is the sample standard deviation of $\mathrm{dIG}^{\mathrm{ctl}}$ over the n_{cal} paired pilot runs (below), which are not reused as test runs. A run is NOT EXERCISED against a control when the on arm never has more candidates than the coalition capacity on a broadcast tick ($|E_k| \le K$ whenever $\chi_k = 1$), so competition never operates; when the on arm has no accessed broadcast; or when $\mathrm{dIG}^{\mathrm{ctl}}_r$ is undefined. Otherwise the run is NEGATIVE if $\mathrm{dIG}^{\mathrm{ctl}}_r \le -\theta^{\mathrm{ctl}}_{\mathrm{eff}}$, WIN if $\mathrm{dIG}^{\mathrm{ctl}}_r \ge \theta^{\mathrm{ctl}}_{\mathrm{eff}}$, and NULL otherwise. The predicted direction is positive.

Decision rule. For each control, the runs classified WIN, NULL, or NEGATIVE enter two one-sided exact sign tests, chosen because they assume only that runs are independent and nothing about the distribution of the contrasts. The WIN test uses the values $u^+_r = \mathrm{dIG}^{\mathrm{ctl}}_r -$

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$\theta^{\text{ctl}}_{\text{eff}}$ and the NEGATIVE test the values $u_r^- = -\theta^{\text{ctl}}_{\text{eff}} - \text{dIG}_r^{\text{ctl}}$. For each test, exactly zero values are dropped, and with $n_\pm$ remaining values of which $n_\pm^{\text{pos}}$ are positive,

$$\text{pv}^\pm = 2^{-n_\pm} \sum_{i=n_\pm^{\text{pos}}}^{n_\pm} \binom{n_\pm}{i},$$

with $\text{pv}^\pm = 1$ when $n_\pm = 0$. The verdict against a control is WIN when $\text{pv}^+ \le \alpha_{\text{test}}$, NEGATIVE when $\text{pv}^- \le \alpha_{\text{test}}$, NOT EXERCISED when every run is NOT EXERCISED, and NULL otherwise. The thesis holds for this form of the architecture when the verdict is WIN against both controls. Requiring both is an intersection-union test, so each control is tested at α_{test} with no multiplicity adjustment. The smallest attainable value is $2^{-n_\pm}$, so at least five runs are needed before any outcome can reach 0.05. With k_n the smallest k for which $2^{-n}\sum_{i=k}^{n}\binom{n}{i} \le \alpha_{\text{test}}$, the power of the WIN test at n runs, when each run's $\text{dIG}_r^{\text{ctl}}$ exceeds $\theta^{\text{ctl}}_{\text{eff}}$ with probability q_{pow}, is

$$\pi(n) = \sum_{i=k_n}^{n} \binom{n}{i} q_{\text{pow}}^i \, (1 - q_{\text{pow}})^{n-i}.$$

Because the test is discrete, $\pi(n)$ is not monotone in n, and the number of runs is set from it as stated below.

Contrast at threshold. In the on arm, a broadcast is accessed exactly when $S_k^* \ge \theta$ (A.1), so the contrast at threshold is a sharp regression discontinuity with running variable S_k^* and cutoff θ (Imbens and Lemieux 2008). Let a broadcast be near threshold when $|S_k^* - \theta| \le \theta_{\text{band}}$. The reports that follow broadcast $\mathcal{B}_k$ are the included reports of every $j \in \mathcal{J}$ whose prediction was formed after j received $\mathcal{B}_k$ and before it received the next broadcast, and each carries the age Δ^b of the context it used. An inhibited broadcast leaves the previous context in place, so the context after it is older by construction; the reports are therefore grouped in age bins fixed with the other settings. For broadcast k and age bin a, let $\bar{g}_{k,a}$ be the mean gain of the reports that follow $\mathcal{B}_k$ in bin a. In run r and bin a, $\hat{\tau}_{r,a}$ is the jump of $\bar{g}_{k,a}$ at $S_k^* = \theta$, estimated by a local-linear fit on each side with a triangular kernel of half-width θ_{band} and corrected for its leading bias by a local-quadratic fit with half-width b_{band} (Calonico, Cattaneo, and Titiunik 2014). A side of bin a can be fitted when, both within θ_{band} and within b_{band} of the threshold, it holds broadcasts with at least n_{side} distinct values of S_k^*, and the weighted least-squares problems of its two fits are well conditioned, the condition number of each, with the score scaled by its bandwidth, at most κ_{max}, so that the fitted coefficients keep about eight significant digits in double precision. A local-polynomial fit behaves as if it had one observation per distinct value of the running variable (Cattaneo, Idrobo, and Titiunik 2024), and with only three distinct values the local-quadratic fit passes through them exactly, so a sparsely populated bin would otherwise weigh as much in the contrast as a well-populated one; $n_{\text{side}} = 10$ is the smallest number of observations those authors recommend on either side of a cutoff, there for the narrowest window of a local randomization analysis. The run's contrast is the mean of $\hat{\tau}_{r,a}$ over the bins in which both sides can be fitted, and it is undefined when there is none. Across runs the contrasts enter a

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one-sided sign test of the same form, on the contrast minus a minimum effect fixed with the other settings. The sign test uses only the spread of the bias-corrected estimates across runs, so it carries the variability of the bias correction, which the robust confidence interval of Calonico, Cattaneo, and Titiunik (2014) accounts for within one sample. That interval, with standard errors clustered by run, is reported for each bin's estimate pooled over the runs.

Speech sound and tone. In a variant of the on arm the tone rule of A.2 is replaced by $I(e) = \Gamma_{\text{Aud}}(\tilde{\nu}_t^{\text{utt}})$ for every tone event, and run r of the variant uses the seeds of run r of the arms. The analysis compares the fraction of emotional tone events that are accessed content with the fraction of neutral ones. The difference of run r is the emotional fraction minus the neutral one, undefined when the run has no emotional or no neutral tone event, and across runs it enters a sign test of the same form. In this variant the intensity depends only on $\tilde{\nu}_t^{\text{utt}}$, so a difference can arise only through the tone model's surprise; the same comparison within bins of $\tilde{\nu}_t^{\text{utt}}$ checks that the ratio accounts for it. The adjusted difference of run r is the mean, over the bins that hold both emotional and neutral tone events, of the difference of the two fractions within the bin, and it is undefined when there is none. It enters a second sign test of the same form, and the bins are not tested separately. Its bins and the minimum effects of its two tests are fixed with the other settings. Both tests are made at α_{test} without multiplicity adjustment, as is the test of the contrast at threshold, because none of them enters the verdict on the thesis.

Settings from the pilot runs. Before the live runs, $n_{\text{cal}} = 30$ paired pilot runs of the three arms and of the tone variant, on seeds not used again, fix every remaining setting by the rules below; thirty is a common rule for the size of a pilot sample that estimates a parameter (Lancaster, Dodd, and Williamson 2004). A pilot run that is NOT EXERCISED is replaced until thirty count, and r_{NE} is the fraction of attempted pilot runs that were NOT EXERCISED. No live run enters any setting.

Learning period. For each arm and each $j \in \mathcal{J}$, the gains of j's reports are averaged across the pilot runs at each report number, and the averages are grouped in consecutive batches of five with means $\bar{Y}_1, \dots, \bar{Y}_b$. The truncation point is that of the marginal standard error rule (White 1997), which a comparison of warm-up methods found the most effective for automatic use (Hoad, Robinson, and Davies 2010),

$$d^* = \arg \min_{0 \le d \le b-5} \frac{1}{(b-d)^2} \sum_{i=d+1}^{b} \left(\bar{Y}_i - \bar{Y}_{(d)}\right)^2,$$

with $\bar{Y}_{(d)}$ the mean of $\bar{Y}_{d+1}, \dots, \bar{Y}_b$. The search leaves at least five batch means after d, as in that comparison, because a few final batch means that happen to lie close together give the statistic a spuriously small value, and at $d = b-1$ it is 0 for every series. T_{learn} is the largest, over arms and processors, of the median entity time at which report $5d^*$ occurs in the pilot runs, rounded up to a whole second, and it applies unchanged to every test run and to the positive control. A truncation point in the second half of a series, $d^* > b/2$, means that the pilot runs are too short, as does a series of fewer than ten batch means, whose search cannot reach the second half. The test does not proceed until longer pilot runs give $b \ge 10$ and $d^* \le b/2$ for every arm and processor.

Minimum effects. $c_{\text{eff}} = 0.5$, half the standard deviation of the contrast, a distribution-based

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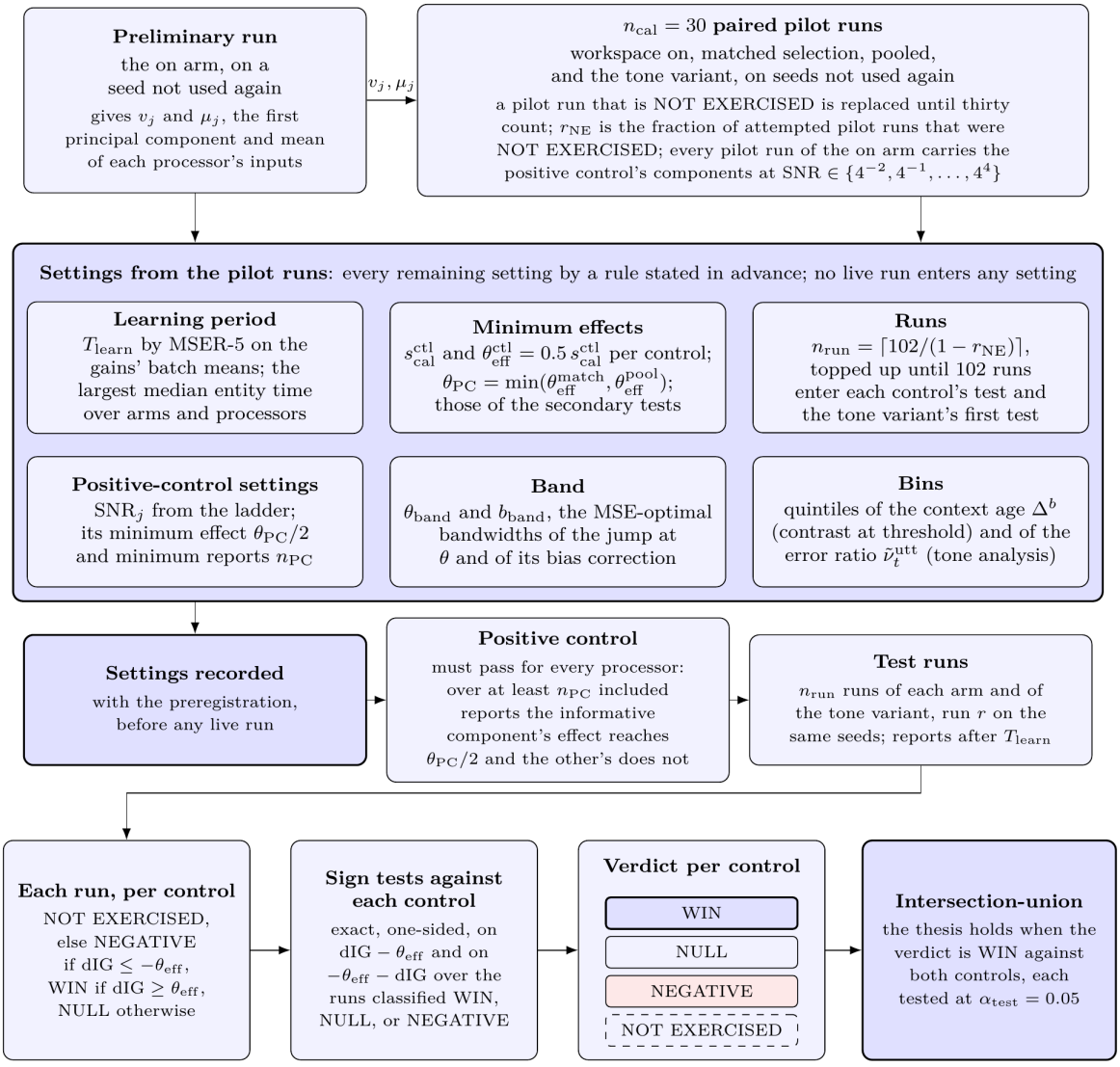


Figure 8: From the pilot runs to the verdict. A preliminary run of the on arm, on a seed not used again, gives each processor's first principal component and input mean. Thirty paired pilot runs of the three arms and the tone variant follow, and a pilot run that is not exercised is replaced. From them every remaining setting follows by a rule stated in advance: the learning period, the minimum effects, the number of runs, the settings of the positive control, the band of the contrast at threshold, and the bins. The settings are recorded with the preregistration before any live run. The test runs start only after the positive control passes for every processor. Each test run is classified against each control, a pair of sign tests gives the verdict against that control, and the thesis holds when the verdict is WIN against both.

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smallest effect of interest (Norman, Sloan, and Wyrwich 2003; Lakens 2017). The minimum effects of the contrast at threshold and of the two tests of the tone analysis are c_{eff} times the standard deviation of the test's run-level statistic over the pilot runs, leaving out runs in which it is undefined.

Runs. $q_{\text{pow}} = 0.65$, Cohen's (1988) medium effect for the sign test. It is a design value and is not estimated from the pilot runs, whose effect estimates are too imprecise to set power (Kraemer et al. 2006). n_{exact} is the smallest n with $\pi(n') \geq 0.9$ for every $n' \geq n$ (Chernick and Liu 2002), which is 102, with $\pi(102) = 0.920$. A power of 0.9 against each control keeps the probability of a WIN against both at or above 0.8, whatever the dependence between the two tests. $n_{\text{run}} = \lceil n_{\text{exact}}/(1-r_{\text{NE}}) \rceil$ runs are planned, and further runs are added until at least n_{exact} runs enter each control's test. Whether a run enters is decided without reference to its contrast, so adding runs on this count leaves the level of the test unchanged, and because $\pi(n) \geq 0.9$ for every $n \geq n_{\text{exact}}$, the final count keeps the planned power. The tone analysis is secondary, but its first test has the power function $\pi(n)$ of the WIN test, so the same q_{pow} and power give it the same n_{exact}. The tone variant is therefore planned with the same n_{run}, and further runs of it are added until at least n_{exact} runs give a defined difference.

Positive control. v_j and μ_j come from the preliminary run. Every pilot run of the on arm carries the two components of each processor at SNR $\in \{4^{-2}, 4^{-1}, \dots, 4^{4}\}$. SNR_j is the smallest of these values at which the informative component's mean effect over the pilot runs reaches θ_{PC}, interpolated linearly in log SNR from the value below it. If the effect stays below θ_{PC} at the largest value, the measure cannot be shown to register an effect of the minimum size, and the live test does not run. With s_j^2 the larger of the two components' variances of the run-level effect over the pilot runs at SNR_j and $\bar{n}_j$ the mean number of included reports of j per pilot run,

$$n_{\text{PC}} = \max_{j \in \mathcal{J}} \left\lceil \frac{(2 z_{0.95})^2 \, \bar{n}_j \, s_j^2}{\theta_{\text{PC}}^2} \cdot \frac{n_{\text{cal}} - 1}{\chi^2_{0.2,\, n_{\text{cal}}-1}} \right\rceil ,$$

where $z_{0.95}$ is the standard normal quantile, so that $2z_{0.95}$ standard errors, the detection limit of Currie (1968) at $\alpha = \beta = 0.05$, equal θ_{PC}. The last factor is the 80 percent upper confidence limit on the variance that Browne (1995) recommends for sizing from a pilot sample. The run-level variance times the reports per run gives the variance per report, and because the control is taken over complete runs, the rule holds whether or not the effect also varies from run to run.

Band and bins. θ_{band} and b_{band} are the bandwidths that minimize the asymptotic mean squared error of the local-linear estimate of the jump at θ and of its local-quadratic bias correction, with a triangular kernel (Calonico, Cattaneo, and Titiunik 2014). They are computed on the pooled pilot broadcasts of the on arm, with one observation per broadcast that has at least one following included report, the mean gain of those reports. The age bins are the quintiles of Δ^b over the reports that follow near-threshold broadcasts in the pilot runs of the on arm, and the tone bins are the quintiles of $\tilde{\nu}_t^{\text{utt}}$ over all tone events, emotional and neutral together, in the pilot runs of the tone variant; five subclasses remove about 90 percent of the bias from a continuous covariate (Cochran 1968). Each bin thus holds a fifth

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of the pilot observations, and the plain mean over the bins gives each bin that share as its weight, the same for both sides of the comparison. This is direct adjustment to the pooled pilot distribution: Cochran (1968) computed the 90 percent for equal weights on subclasses of equal size, and Rosenbaum and Rubin (1984) used the same weights with quintiles of the pooled sample.

A.9 Multi-seed stability

For per-seed headline values $Y_1, \dots, Y_{n_{\text{seed}}}$, let $\bar{Y}$ be their mean and σ_Y their population standard deviation ($\sigma_Y = 0$ for a single seed). The coefficient of variation is

$$\text{CV} = \begin{cases} \sigma_Y / |\bar{Y}| & \bar{Y} \neq 0, \\ 0 & \bar{Y} = 0,\ \sigma_Y = 0, \\ \infty & \bar{Y} = 0,\ \sigma_Y > 0. \end{cases}$$

An ensemble is stable when $\text{CV} \leq \text{tol}_{\text{CV}}$ and at most one distinct verdict outcome occurs across seeds; an ensemble without verdicts counts as unanimous, and an infinite CV is never within tolerance. In the offline suite the check runs on the ablation of the coherence layer with n_{seed} seeds. The criterion measures run-to-run spread and verdict agreement; dynamical stability within a run is outside its scope, and the coefficient of variation is ill-conditioned when the mean is near zero.

A.10 Memory strength and sleep

The episodic-memory module is held in the base-thesis form; this subsection states the rules of §4 that apply when it is present. The module stores one trace for each broadcast. Each trace i in its persisted stores has a strength $h_i \in (0, 1]$, set to 1 when the trace is stored and changed only by the downscaling below, and a storage time read from the entity clock. Let $\mathcal{W}$ be the set of traces stored since the previous downscaling, and $N_{\text{new}} = |\mathcal{W}|$.

Replay. In each sleep the module first replays the set $\mathcal{Q}$ of the n_{rep} traces of $\mathcal{W}$ with the highest score $c_{\text{aff}}\, a_i^{\text{enc}} + c_{\text{rec}}\, \text{rec}_i$, or all of $\mathcal{W}$ when it has fewer. Here a_i^{enc} is the last arousal (A.4) that the affect module published before trace i was stored (0 if none), and rec_i maps the trace's storage time linearly onto $[0, 1]$ across $\mathcal{W}$, 1 for the newest trace and 0 for the oldest; when all share one storage time, $\text{rec}_i = 1$.

Downscaling. After the replay every trace outside $\mathcal{Q}$ is scaled by one factor,

$$h_i \leftarrow \varrho^{\,N_{\text{new}}/N_{\text{day}}}\, h_i, \qquad i \notin \mathcal{Q},$$

and the traces in $\mathcal{Q}$ keep their strengths. $N_{\text{day}} = 86\,400\, f_0$ is the number of traces that one entity day at the resting broadcast rate stores, so a trace that is never replayed keeps the fraction ϱ of its strength over each such day of storage. The factor is common to every scaled trace, so the ratio of any two of their strengths is unchanged, and a sleep after a wake that stored nothing changes no strength. With storage at the resting rate the total strength $\sum_i h_i$ approaches $N_{\text{day}} / \ln(1/\varrho)$, about five days of storage, and from then on each sleep returns

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the total to about its level before the preceding wake, the renormalization to a baseline that the hypothesis describes.

Pruning. After the downscaling, in the same sleep, the module deletes every trace with $h_i < h_{\min}$. Then, while the size S of the persisted stores exceeds the budget B, it deletes the trace outside $\mathcal{Q}$ with the lowest strength, the oldest first among equal strengths, and it stops when $S \le B$ or no trace outside $\mathcal{Q}$ remains. S is the sum, over the persisted traces, of the size in bytes of each trace's stored record, its embedding together with its payload. B is set by the operator from the disk the installation can give the stores, so it belongs to the installation, and the model gives it no value. The traces in $\mathcal{Q}$ keep strength 1, so the floor never reaches them, and the budget passes over them. Deletion happens only here, so the stores hold no trace below the floor while the entity is awake, and between two sleeps S can exceed B by the records stored since the last sleep, about 12,000 traces per entity hour at the resting rate and at most 36,000 at the ceiling of one broadcast per tick. A trace that is never replayed reaches the floor after about $\ln h_{\min} / \ln \varrho$ entity days of storage at the resting rate, about 23 days with the values of Table A1, so with storage at the resting rate and a budget that does not bind, the stores settle at about $N_{\text{day}} \ln h_{\min} / \ln \varrho$ traces, about 6.7 million. A preservation copies the stores and the strengths as they stand at capture, so a later pruning deletes nothing from a preserved entity. Each sleep records N_{new}, the factor, $|\mathcal{Q}|$, the total strength before and after the scaling, the numbers of traces deleted at the floor and for the budget, and S after the pruning.

Recall. For a cue, let sim_i be the cosine similarity between the embeddings of the cue and of trace i. Recall returns the requested number of traces with $\text{sim}_i > 0$, in decreasing order of $h_i \, \text{sim}_i$, the inner product of the cue's unit embedding with the trace's unit embedding scaled by h_i. A trace in the short-term buffer has strength 1 and ranks by similarity alone. Recall changes no strength.

A.11 Parameter values

Table A1. Parameter values. "Design" marks a value that is part of the model's definition; "calibrated" marks a value set offline by simulation or by the validation of A.7; "provisional" marks a current setting to be calibrated before the live runs, when the values used are recorded with them; "operator" marks a setting that each installation chooses for its own resources. Times marked "entity" are on the entity clock, "awake" is awake time (A.7), and "wall" is wall-clock time.

Symbol	Meaning	Value	Units	Status
f_p	processing rate	10	Hz (entity)	design
f_0	resting broadcast rate	10/3	Hz (entity)	design
none	entity seconds per wall-clock second	1	none	design
B	maximum events read per stream per tick	100	events	design
K	coalition capacity	5	events	design
θ	access threshold	0.35	none	provisional
W_N	novelty window	32	candidates	design
G	goal factor (static)	1	none	design

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Symbol	Meaning	Value	Units	Status
γ_G	attenuation of the inactive drive-relevance goal factor	0.5	none	design
T_{min}, T_{max}	arousal level-gain floor and ceiling	0.2, 1.0	none	design
g_{max}	arousal contrast gain at arousal 1	8.0	none	design
none	coherence layer	off	none	design
κ_{min}, κ_{max}	coherence factor floor and ceiling (layer on)	0.8, 1.25	none	design
W_κ	coherence phase window (layer on)	10	ticks	design
$I^{\text{lo}}_{\text{Vis}}$, $I^{\text{hi}}_{\text{Vis}}$	vision intensity levels	0.2, 0.7	none	provisional
$I^{\text{lo}}_{\text{Aud}}$, $I^{\text{hi}}_{\text{Aud}}$	audition intensity levels	0.4, 0.8	none	provisional
$I^{\text{lo}}_{\text{Intero}}$, $I^{\text{hi}}_{\text{Intero}}$	interoception intensity levels	0.1, 0.7	none	provisional
$I^{\text{lo}}_{\text{Seq}}$, $I^{\text{hi}}_{\text{Seq}}$	sequence intensity levels	0.1, 0.7	none	provisional
$I^{\text{lo}}_{\text{Aff}}$, $I^{\text{hi}}_{\text{Aff}}$	affect intensity levels	0.1, 0.7	none	provisional
$I^{\text{lo}}_{\text{Sleep}}$, $I^{\text{hi}}_{\text{Sleep}}$	sleep intensity levels	0.5, 0.8	none	provisional
$I^{\text{lo}}_{\text{Lang}}$	language-organ utterance intensity	0.4	none	provisional
β_Γ	error ratio at which the graded map reaches the alert level	2	none	design
W_P	running-ratio window (vision, audition, interoception, sequence)	32	reports	design
β_ν	prediction-error alert ratio (vision, audition)	2.0	none	design
β_c	change alert ratio (vision, audition)	2.0	none	design
$c^{\text{min}}_{\text{Vis}}$, $c^{\text{min}}_{\text{Aud}}$	absolute change floors (vision, audition)	10^{-4}, 0.35	none	design
W_{buf}	forward-model input buffer (vision, audition)	16	reports	design
none	forward-model hidden units (vision; audition)	256; 32	units	design
η_{fm}	forward-model learning rate (vision, audition, interoception, sequence)	10^{-3}	none	design
ω_{min}, ω_{max}	audition attended fraction at arousal 1 and 0	0.15, 1.0	none	design
Δ_{att}	shortest attended span (audition)	0.025	s (entity)	design
none	vision saliency grid	12 by 12	tiles	design
Θ_F	fovea hysteresis margin	0.15	none	design
F_{min}, F_{max}	fovea size at arousal 1 and 0	0.12, 0.5	fraction of frame	design

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Symbol	Meaning	Value	Units	Status
α_F	fovea tile-statistics weight	0.05	none	design
n_F	tile updates before z-scoring	5	clip reports	design
k_F	tile deviation floor, as a fraction of the mean tile change	0.1	none	design
β_ν^{Seq}	sequence temporal-error alert ratio	3.0	none	design
W_z	sequence hidden-state z-score window	64	broadcasts	design
W_{recur}	recurrence window	32	broadcasts	design
n_{recur}	recurrence count	4	states	design
Δ_{recur}	recurrence quantization step	0.25	none	design
$W_{\Delta t}$	sequence interval window	32	broadcasts	design
s_{max}	timespan cap (interoception, sequence) and cap on an entered interval relative to the window mean (sequence)	10	none	design
none	reservoir units (interoception; sequence)	32; 32	units	design
none	sequence forward prediction	on	none	design
I_0	phasic salience floor	0.5	none	design
τ_{ph}	phasic decay time	1.0	s (entity)	design
a_0	baseline arousal (affect and access rate)	0.3	none	design
none	adaptive access rate	on	none	design
λ	arousal relaxation rate	0.05	s^{-1} (entity)	design
P_{Aff}	minimum interval between affect state events, and timer period	1.0	s (entity)	design
γ_a	perceptual-alert arousal gain	0.15	none	design
ν_{cap}	cap on the excess error ratio per alert	4.0	none	design
γ_{Intero}	arousal step per interoception hard-threshold report	0.05	none	design
τ_{f}, τ_{s}	fast and slow time constants of the error means and alert rates	10, 100	s (entity)	design
k_v	learning-progress gain in the pleasantness check	2.0	none	design
τ_v	valence time constant	30	s (entity)	design
lp_0	learning-progress noise floor	0.05	none	calibrated
m_B	alert-excess margin	0.5	none	calibrated
r_0	alert-rate floor	0.05	s^{-1} (entity)	design
α_{int}	intent-rate weight	0.05	none	design
β_D, δ_D, ρ_D	curiosity build, decay, relief	0.05, 0.0055, 0.5	s^{-1} (entity)	design

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Symbol	Meaning	Value	Units	Status
β_D, δ_D, ρ_D	boredom build, decay, relief	0.04, 0.0045, 0.3	s^{-1} (entity)	design
β_D, δ_D; ρ_D	social-drive build, decay; relief per interaction	0.01, 0.0011; 0.8	s^{-1} (entity); none	design
β_D, δ_D; ρ_D	restlessness build, decay; relief per intent	0.03, 0.0033; 0.5	s^{-1} (entity); none	design
θ_D	drive threshold (every drive)	0.7	none	design
Θ_D	drive re-arm fraction	0.9	none	design
none	initial held arousal $\hat{a}$	0.3	none	design
none	wellness scaling: GPU temperature bounds; cycle-latency target	30 to 80; 300	°C; ms	design
P_{Intero}	interoception reading interval	1.0	s (entity)	design
n_x	interoception feature dimension (three self-rhythm features are zero unless the self-rhythm is enabled, and one is the highest GPU memory use)	8	none	design
τ_U	expected-error time constant	600	s (entity)	design
β_U	expected-error band	2.0	none	design
none	hard thresholds: CPU, RAM, GPU temperature, GPU memory, cycle latency	90, 90, 83, 92, 600	%, %, °C, %, ms	design
θ_R	regulation threshold	0.5	none	provisional
Δ_R	regulation sustain window	30	s (entity)	design
n_{wu}, Δ_{wu}	warm-up end conditions	1000, 1200	updates, s (entity)	design
γ_f	processing-rate reduction factor	0.8	none	design
$f_p^{\min}$, $f_p^{\max}$	processing-rate bounds under regulation	0.5, 20	Hz (entity)	design
θ_{th}	think bar	0.45	none	provisional
θ_{sp}	speak (report) bar	0.6	none	provisional
Δ_{th}	think refractory interval	3.0	s (entity)	design
Δ_{sp}	speak refractory interval	8.0	s (entity)	design
Δ_{sig}	signature expiry	300	s (entity)	design
Δ_g	in-flight guard timeout	48	s (wall)	design
none	interrupt bar	unset	none	design
f_r	self-rhythm step rate	20	Hz (entity)	design
n_s	Euler substeps per step	10	none	design
J_{rec}	recurrent weight	2.5	none	calibrated
Ξ_0	input offset	-0.25	none	calibrated
k_y	sigmoid slope scale	0.08	none	calibrated
τ_y	activity time constant	0.02	s (entity)	calibrated

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Symbol	Meaning	Value	Units	Status
γ_R	depression rate	8.0	s^{-1} (entity)	calibrated
γ_o, γ_u	own-drive and maternal-drive gains	0.02, 0.03	none	calibrated
σ_ξ	noise scale	0.02	none	calibrated
$\tau_{\text{rec}}^{(0)}$	initial recovery time	2.1	s (entity)	calibrated
$\tau_{\text{rec}}^{\text{min}}, \tau_{\text{rec}}^{\text{max}}$	recovery-time bounds	0.9, 2.3	s (entity)	calibrated
η_ℓ	frequency-adaptation rate	0.0025	s^{-1} (entity)	calibrated
τ_m	moving-average time constant	10	s (entity)	design
Δ_{amp}	phase and amplitude window	8	s (entity)	design
none	initial activity y and resource R	0.05, 1.0	none	design
none	maternal beat rate and drift	70, 0.03	bpm, none	design
M_{max}	maximum maternal drive	0.4	none	design
$c_M^{\text{base}}, c_M^{\text{pert}}$	usual and perturbation drive scales	0.5, 0.75	none	design
f_g	readout sampling rate	10	Hz (entity)	design
$f_{\text{lo}}, f_{\text{hi}}$	entrainment band	0.3, 2.0	Hz (entity)	design
Δ_{trim}	edge trim	2.0	s (entity)	design
Δ_E	driven-window length	300	s (entity)	design
Δ_W, P_W	withdrawal duration and period	20, 1800	s (entity)	design
$\Delta_{\text{pert}}, P_{\text{pert}}$	perturbation duration and period	5, 3600	s (entity)	design
j_P	probe jitter fraction	0.25	none	design
P_{ro}	readout period, also the settling time after boot or thaw	60	s (entity)	design
f_{guard}	pull guard	0.05	Hz (entity)	design
pull_{min}	pull floor	0.5	none	calibrated
n_{base}	baseline withdrawals	3	count	design
none	surrogate heartbeats	19	count	design
none	consecutive passes required	3	count	calibrated
$t_{R0}, t_{R1}, t_{R2}, t_{R3}$	viability checkpoints	6, 24, 48, 60	h (awake)	calibrated
pull_{R1}, slope_{R1}	R1 bounds on median pull and slope	0.12, 0.002	none, h^{-1} (awake)	calibrated
pull_{R2}	R2 bound on median pull	0.3	none	calibrated
Δ_V, n_V	viability window and minimum points	12, 8	h (awake), withdrawals	calibrated
Δ_{per}	period-variability window	300	s (entity)	design
$V_{\text{per}}^{\text{min}}$	period-variability floor	0.2	none	provisional
$\text{err}_{\text{Vis}}^{\text{max}}$	vision error-ratio ceiling	0.3	none	provisional
$\Delta_{\text{max}}^{\text{rec}}$	return-to-baseline ceiling	30	s (entity)	provisional
$\text{tol}_{\text{rec}}, \Delta_{\text{cap}}^{\text{rec}}$	recovery tolerance and cap	0.25, 300	none, s (entity)	design

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Symbol	Meaning	Value	Units	Status
n_{sleep}, n_{cons}	required sleeps and consolidation passes	5, 3	counts	design
Δ_{awake}	minimum awake time	24	h (awake)	design
none	gate cadence	60	s (entity)	design
Δ_{budget}	developmental-phase budget	96	h (wall)	design
c_{eff}, n_{cal}	minimum-effect multiple and paired pilot runs	0.5, 30	none, runs	design
n_{run}	runs per arm and of the tone variant	$\lceil 102/(1 - r_{\text{NE}}) \rceil$ planned, at least 102 entering each control's test and the tone variant's first test (A.8)	runs	design
q_{pow}	assumed probability that the contrast exceeds the minimum effect (power analysis)	0.65	none	design
none	power of each control's WIN test	0.9	none	design
T_{learn}	initial learning period excluded from each run and from the positive control	from the pilot runs (A.8)	s (entity)	design
v_j, μ_j	direction and mean of the positive control's projection	from a preliminary run (A.8)	none	design
SNR_j	signal-to-noise ratio of the positive control's component	from the pilot runs (A.8)	none	design
$\theta_{\text{PC}}/2$	minimum effect of the positive control	from the pilot runs (A.8)	none	design
none	detection probabilities of the positive control (informative, uninformative)	0.95, 0.05	none	design
n_{PC}	minimum included reports of the positive control	from the pilot runs (A.8)	reports	design
θ_{band}, b_{band}	half-width of the threshold band and bandwidth of its bias correction	from the pilot runs (A.8)	none, none	design

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Symbol	Meaning	Value	Units	Status
none	context-age bins of the contrastive analysis	quintiles, from the pilot runs (A.8)	s (entity)	design
n_{side}	distinct scores required on each side of a fitted bin of the contrast at threshold	10	broadcasts	design
$\kappa_{\max}$	largest condition number of a fitted side's weighted least-squares matrix	10^{8}	none	design
none	error-ratio bins of the tone analysis	quintiles, from the pilot runs (A.8)	none	design
none	minimum effects of the contrast at threshold and of the tone analysis	0.5 times the pilot standard deviation (A.8)	none	design
α_{test}	level of each sign test of the live test	0.05	none	design
α_{FW}	family-wise level (offline suite)	0.05	none	design
tol_{CV}, n_{seed}	stability tolerance and seeds (offline suite, coherence-layer ablation)	0.05, 3	none, seeds	design
ϱ	strength kept by a trace not replayed over one entity day of storage at the resting rate (A.10)	0.82	none	design
N_{day}	traces stored by one entity day at the resting broadcast rate, $86\,400\,f_0$ (A.10)	288,000	traces	design
$h_{\min}$	pruning floor of trace strength (A.10)	0.01	none	design
B	storage budget of the episodic-memory stores (A.10)	set by the operator	bytes	operator
n_{rep}, c_{aff}, c_{rec}	traces replayed per sleep and weights of the replay score	5, 0.7, 0.3	traces, none, none	design

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